1.1. The project summary

Project Number [1]	958402	Project Acronym [2]	AI-CUBE

One form per project	
General information	
Project title [3]	Artificial Intelligence and Big Data CSA for Process Industry Users, Business Development and Exploitation
Starting date [4]	01/09/2020
Duration in months [5]	24
Call (part) identifier [6]	H2020-NMBP-ST-IND-2020-singlestage
Topic	DT-SPIRE-11-2020 Artificial Intelligence and Big Data Technologies for Process Industries (CSA)
Fixed EC Keywords	Artificial intelligence, intelligent systems, multi agent systems, Smart manufacturing
Free keywords	

Abstract [7]

AI-CUBE seeks to enhance the understanding of different digital technologies related to artificial intelligence (AI) and big data (BD) applied in process industries for all the SPIRE industrial sectors (cement, ceramics, chemicals, engineering, minerals and ores, non-ferrous metals, steel, water). Therefore, a close collaboration with industry is mandatory to achieve in-depth insights into possible application areas of AI for processes, technology, sensor applicability and assessment of their level of penetration.

The overall project approach is based on the development of a 3-dimensional conceptual matrix based on:

1) AI and BG technologies
2) Application areas (activities and industrial processes)
3) SPIRE sectors

AI-CUBE's main goal is to define a roadmap in AI and the use of BD for the process industry and their maturity level across the industrial sectors, including guidelines for implementation. Industrial stakeholders and associations will validate the consolidated roadmap ensuring solution feasibility and benefits for the European industrial community. A crosslinked vision over process industry sectors shall facilitate cooperation and boost technologies deployment at their full potential. An in-depth consultation with industry (association, representatives, companies) will provide an overview of current AI and BD algorithms application, identifying exploitable synergies among sectors. A deep study of the application areas in planning and operations within other industrial sectors facilitates a gap analysis, propitiating knowledge sharing among processes and sectors.

A Multi-Actor Multi-Criteria analysis will obtain a widely supported and consensus-based action plan for industrial consultation. This will allow the inclusion of a broad stakeholder community representing the main industry actors throughout all the SPIRE sectors, with which the project consortium has strong connections that will support sector integration and stakeholders' engagement.

1.2. List of Beneficiaries

Project Number [1]	958402	Project Acronym [2]	AI-CUBE

List of Beneficiaries

No	Name	Short name	Country	Project entry month[8]	Project exit month
1	CIAOTECH Srl	PNO	Italy	1	24
2	FUNDACION ZARAGOZA LOGISTICS CENTER	ZLC	Spain	1	24
3	FRAUNHOFER GESELLSCHAFT ZUR FOERDERUNG DER ANGEWANDTEN FORSCHUNG E.V.	Fraunhofer	Germany	1	24
4	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	IRIS	Spain	1	24
5	CONSIGLIO NAZIONALE DELLE RICERCHE	CNR	Italy	1	24

1. Excellence

The European Sustainable Process Industries community, especially through A.SPIRE is in a unique position to facilitate, speed-up and harmonise progress in digitisation, artificial intelligence (AI) and big data (BD) technologies across different process industries, from chemicals to steel, cement, and ceramics, up to water and engineering. AI and big-data technologies are developing fast, but in a **fragmented** and diverse manner **throughout the different process industries sectors** and **across different organisational functions and processes** within the companies.

On one hand, AI and BD technologies seem further advanced in analytics and research domains[1] (e.g. for simulations in material science), which are less critical to the core business process. On the other hand – depending on the process industry sector – more critical AI and BD supported organisational applications are emerging and becoming increasingly essential to drive production processes towards their optimum. Some examples are model predicted process control and optimisation and, in some cases, digital process control of remotely connected plants of the same organisations (e.g. process intensification in the chemical industry) or predictive (tele-) maintenance based on an increasing amount of sensor data combined with digital twins. These applications use increasingly lower and changing feedstock qualities to produce output materials within ever tighter specs, while optimising the energy and resources consumption. (Mass) customisation trends, exacerbate this trend. New Supply Chain initiatives, targeting for instance industrial symbiosis, are moving to the
'platformitisation' of e.g., resource management, driven by digitisation, offering further opportunities for AI and BD applications that cross the boundaries of various process, manufacturing and logistics industries.
Cutting through these developments, the trend of "**modularization of the industrial value chain**" emerges. As already spotted in a publication by AT Kearney[2] almost twenty year ago, and recently further analysed and elaborated for the process industry in the **A.SPIRE CSA "INSPIRE**"[3] on future business models, there is a global trend towards what they defined "remodularization of the corporate genome". It observes a trend towards capability-driven organisations that will be more transparent, agile and manageable. It predicts that there will increasingly be a split between single-capability approaches and business propositions built on recombining digitally supported capabilities along the value chain from raw materials towards end-products and recycling. This overall trend may have significant impact on reconfiguring full value chains across borders, where each value chain function is carried out by organisations that performs best, as such stimulating new business models and side-streams in the value chain e.g., to do maintenance, deliver on-demand intermediate materials, products or process components (flexible customisation), services instead or on top of products (servitisation), including new businesses and services to optimise resource use and reuse along the value chain (optimisation and industrial symbiosis). AI and BD are expected to play an ever more important enabling role in such a modularised process industry value chain, where collaborations may be continuously reconfigured within virtual spaces, aided by AI-based prediction and decision support systems.

These trends towards a more modularised, agile and flexible digitally enabled process industry, represented in the following figure generated within the INSPIRE project, establishes the vision and background of the AICUBE consortium, against which it will assess the use and the potential of AI and BD technologies to support the process industries of the future.

[1] Deloitte Insights, *State of AI in Enterprise*, 2nd Edition, 2018.
[2] Rebuilding the corporate genome, Johan C. Aurik, Gillis J.Jonk, Robert E. Willen, 2002
[3] H2020-IND-CE-2016-17: INSPIRE project "Towards growth for business by flexible processing in customer-driven value chains" 2017 – 2019.

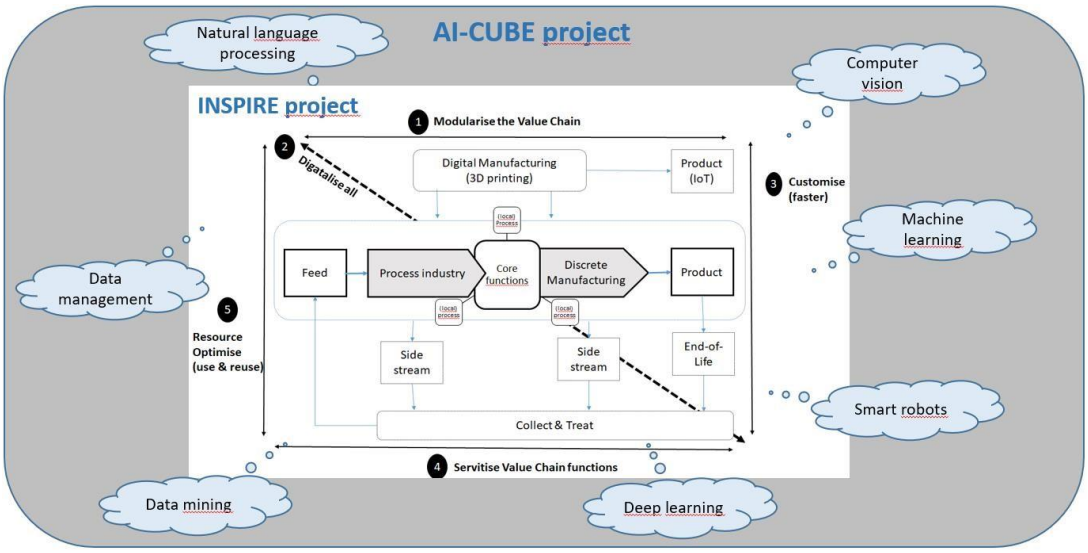


Figure 1. Development trends of digitally enabled process industry (Source: INSPIRE Project)

1.1 Objectives

Digitisation, AI and BD technologies, are currently being developed, tried and deployed differently, at different maturity levels between the different process industries and organisational functions. This represents both a challenge and an opportunity for the European process industry. A **challenge** because the benefits of AI and BD solutions to optimise the process towards the SPIRE vision of sustainable and competitive process industries are not equally being realised in all the process industry sectors, or processes. At the same time, this also offers an **opportunity**, as positive experiences with AI and BD in one process industry sector can be adapted and transferred to another, once it has become clear what their benefits are and how further development towards AI and big data business cases can be pursued.

The AI-CUBE project aims to tackle this challenge and leverage on this opportunity, by identifying the use and penetration of AI and BD in each of the process industries and organisational processes, as a basis for cross-sectorial knowledge and technology transfer and to design business case-oriented roadmaps for the European process industries of the future.

The overall objective of AI-CUBE project is to contribute to **harnessing and optimising the potential of AI and BD in the European process industry**, by defining a roadmap that serves as guidance for researchers, managers and operators, with specific recommendations for all involved industrial sectors and organisations' functions and processes. To achieve this, the specific objectives of AI-CUBE are:

1. draw a 4-dimensional **AI and BD map (or a "CUBE")** where good practices are available within the different organisational processes across the eight SPIRE process industries, assessing the **current state-of-play and level of penetration of AI and BD** within the different organisational processes across the 8 SPIRE process industry sectors;
2. identify **good practices** to build on towards **future AI and BD business cases** and define the RD&I actions needed in a roadmap per sector;
3. detect the **white spots of AI and BD solutions**, that can be covered by good practices from other process industries, and outline the roadmap towards adapting good practices for other sectors;
4. define the **data requirements, skills and RD&I requirements** for future AI and BD business cases to emerge within the different process industry sectors.

All these will converge into **8 AI and BD roadmaps**, one for each SPIRE sector, indicating a route map and practical recommendations on AI and BD business cases, transferability of good practices from other industries, data, skills and RD&I requirements, in line with the A.SPIRE 2050 vision.

The innovative approach proposed in AI-CUBE, hence, works on **3 axes**: **1)** the different AI and BD technologies, **2)** the eight SPIRE process industries, and **3)** the different organisational processes where AI and BD will make the difference in the future, combined with a Maturity Level Assessment framework.

Ad. 1: AI and BD technologies AI-CUBE will make a selection of the most relevant AI and BD technologies in the first phase of the project, starting from what is currently typically appreciated by most scientists, industry and ICT managers as the most important in the deployment of AI and BD solutions, such as the AI technologies visualised in fig. 2[4].

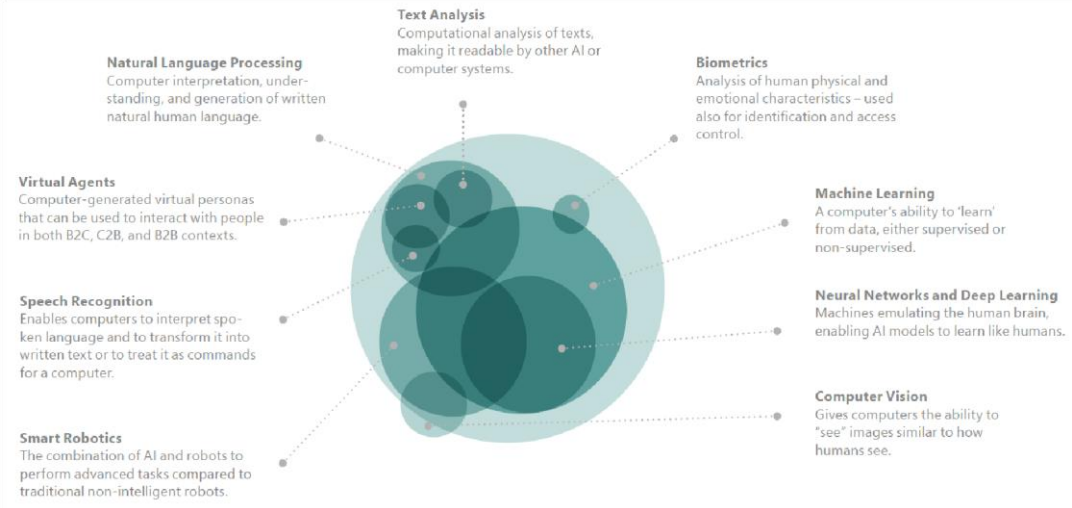


Figure 2. Artificial intelligence and big data technology deployed in Europe

77% of industry managers considered machine learning to the be most useful in deploying AI, with smart robotics second (44%), natural language processing third (40%). Key big-data technologies such as datastorage, mining, analytics, visualisation etc., will also be considered. AI-CUBE will carry out a specific stocktaking exercise to define the most relevant AI and BD technologies for the future of the process industry, as input for its initial mapping. Previous and ongoing research, development and innovation projects from H2020, several national funding programmes in Europe, the US and Canada, in the domain of AI and BD for the process and manufacturing industries (see under ad. 2) as well as the different organisational core processes (see ad. 3) will be mapped and analysed, as a basis to scope the mapping exercise. A preliminary table of relevant projects has already been set-up (see Table 1 in section 1.3.4, Relation to previous initiatives).

Ad. 2: Process Industries. All eight A.SPIRE process industry sector will be considered for the mapping. Additionally, in our study, also Pulp and Paper will be taken into consideration and – in close alignment with A.SPIRE – further process industries may be included (e.g. the glass-industry), or specific manufacturing industries that can serve as inspiration or basis for AI and BD technology transfer to the process industries (e.g. automotive).

Ad. 3. Organisational core-processes: AI-CUBE will map the state-of-play of AI and BD in different organisational core-processes, especially where AI and BD are (expected) to play a key role in the future, taking into consideration the macro-areas mentioned in the call text (e.g. RD&I, process control, SC management, predictive maintenance and product customisation and traceability). Such mapping will furthermore facilitate comparison between AI and BD technology developments and maturity, the different process industry sectors, and reasons about transferability.

For each AI and BD, a **Maturity Levels Assessments** (**MLA**) will be carried out, based on an assessment method specifically designed by the AI-CUBE partners for the project, which will combine a selection of the state of the art methodologies to "measure" maturity in five AI and BD development phases: inception, elaboration, construction, deployment and maintenance[5]. The model can be used for three purposes[6]:

[4] Artificial Intelligence in Europe: An outlook for 2019 and Beyond – EY for Microsoft

[5] Garcia-Mireles, G.A., Moraga, M.A., Garcia, F.: Development of maturity models: a systematic literature review. In: 16th International Conference on Evaluation Assessment in Software Engineering (EASE 2012), pp. 279–283 (2012) [6] De Carolis, A., Macchi, M., Negri, E., & Terzi, S. (2017, September). A maturity model for assessing the digital readiness

1. to assess the as-is situation of the organization/process;
2. to approach maturity improvement in order to positively affect business value;
3. to enable benchmarking across companies; in particular, a model of this nature would be able to compare similar practices across organizations

Maturity levels will be mapped in the AI-CUBE, in order to enable comparative analysis between states of development, transferability and RD&I needs towards future AI and BD business cases to be defined within the AI-CUBE project.

This approach, which will underpin all actions of the AI-CUBE CSA project, can be visualised in fig. 3, where the four dimensions are represented 1) AI & BD technologies, 2) SPIRE sectors, 3) Organisational coreprocesses where AI & BD can make a difference, and 4) AI & BD Maturity Levels.

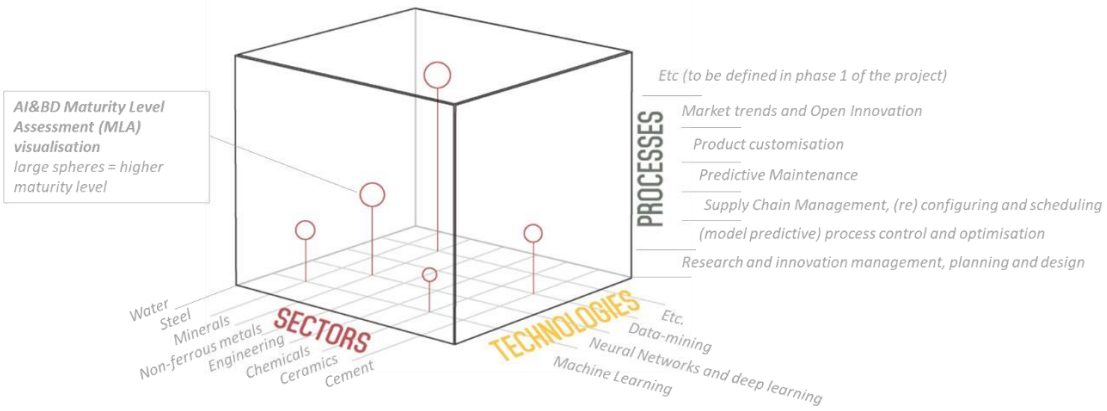


Figure 3. The "cube" concept

Once the current STATUS of AI and BD use has been "cubed", future AI and BD driven business cases for the European process industries will be defined together with industry stakeholders, several of whom have already indicated support (see Letters of Support), and a gap analysis will be carried out to define required skills, data and RD&I within a set of roadmaps (see: 1.3.1 Concept and approach).

1.1.1 Challenges of the AI technologies and use of big data in the industry

In the era of digitalization, industries are faced with new challenges. Indeed, to fully exploit the potential of data driven technologies, the process industry should be aware of inherent challenges and technical barriers.

Some AI methods are based on large amounts of data and this data must be highly reliable, therefore data generation (e.g. by sensors), data robustness, data quality, and data access need to be improved further in process industry. In many cases, process industries do not have enough data available to apply machine learning techniques in a suitable way. New technologies must be developed which are able to work with a significantly lower amount of data to produce similar results.

Big Data controls many aspects of modern society and transforms how companies from all industries can create value when data is collected and analysed. It is defined by the rapid growth of digital data volumes that can be generated from **different sources** in **different data structures** by an increasing number of users, sensors, processes and other sources[6]. Big data is mostly collected beyond the traditional database software

of manufacturing companies. In *IFIP International Conference on Advances in Production Management Systems* (pp. 13-20). Springer, Cham.

[6] Wolf, Oliver; Leveling, Jens; Luciano, Francesco. Big Data: große Datenmengen der Logistik beherrschen und nutzen, Zeitschr.-Verlag.Breidenstein, Frankfurt am Main, 2015.

tools to enable access, storage, management and analysis of large amount of data[7]. In many cases it is thus a prerequisite for the use of AI[8]. Process industry already generates a vast amount of data within its general processes. The challenge remains to integrate the data and define data structures and interfaces to create a common access over several processes and locations.

1.1.2 Rising the challenges

The project will identify good practices and potential of AI and BD within a digitalized process industry, and guide stakeholders towards defining development, take-up and implementation routes for the most promising data driven technologies (AI and big data) and business cases. It will help making them accessible to the involved actors in the industry and to rise the data driven challenges by:

Assessing and mapping ("cubing") the most important **AI technologies and use of big data** in different macro-application areas in the SPIRE process industry sectors (including level of implementation and maturity of data driven technologies based on a new AI and BD Maturity Level Assessment Model to harmonize assessments across process industries).
Identification of **AI and BD potential through future oriented business cases**, and define gaps within the different SPIRE process industries, including the potential for synergies between the different process industries for technology transfer, adaptation and deployment of AI and BD solutions.
Definition of a **roadmap for the SPIRE process industry** as a guideline for the identification of AI and BD opportunity spotting and implementation routes towards promising business cases of the future.

The AI and BD Business Cases of the future will be selected through direct industry involvement, considering both their impact on global competitiveness of the European process industries and the contribution to the new SPIRE 2050 vision and the new EC's "Green Deal" policy. This project, hence, aims to shed light on the potential of AI and BD technologies to transform the process industry in the new era of Industry 4.0, as well as the targeted climate neutrality, as expressed by the European Commission's Green Deal.

1.2 Relation to the work programme

Both the objectives and the expected impact of AI-CUBE are fully in line with the challenges and scope of the H2020 topic SPIRE-11-2022: Artificial Intelligence and Big Data Technologies for Process Industries.

SPIRE-11-2020	AI-CUBE
Identification, based on a mapping of digital technologies in process industries and their level of penetration, which specific AI- and big data technologies are most relevant, and what are, or could be, the most relevant application cases and/or pilots in process industry.	WP3, receiving input from WP1 and WP2 will develop in Task 3.2 a mapping on AI and BD technologies per sector and macro area, as well as a cross-comparison of the researched AI and BD technologies and the fields of application in different process industries. An assessment of the level of penetration / maturity degree of the different AI and BD technologies will also be performed, identifying the technology application to each industry and AI technology combination.
A roadmap should be developed for all the different sectors in the process industries to take full advantage of AI and big data and give clear and pragmatic recommendations for researchers, managers, and operators planning to harness their potential.	AI-CUBE roadmap in AI and BD technologies will be developed in WP4 (Task 4.3). This roadmap will be a guideline in the implementation of these technologies of application in process industries for all the SPIRE industrial sectors with specific recommendations to guide researchers, managers and operators in the implantation of these technologies. The roadmap will allow all the SPIRE sectors to benefit and share the knowledge gained to boost the competitiveness of the European industry. Industrial stakeholders and associations will validate the process (WP2) and accompany the roadmap analysis.

[7] Wang, Yaxing et al. Research on the Impact of Big Data on Logistics, Zhengzhou University, China, 2017
[8] Rügheimer, Hannes. Lässt sich ein Computer in die Irre führen? https://intelligente-welt.de/sicherheit-machine-learning/, 2017

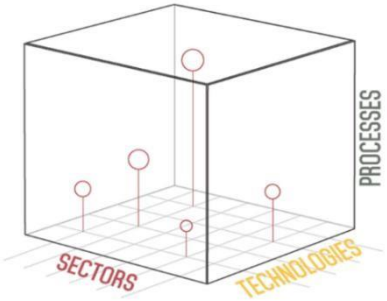
The roadmap should explore, inter alia, actions on: • Research and innovation management, planning, and design (e.g. new chemical synthesis strategies, health and safety assessments) • Process control: yield and accuracy enhancement • Supply chain management and scheduling of connected processes, plants and/or sites (e.g. for industrial symbiosis), process flexibility • Predictive maintenance • Product customisation and product traceability	WP1 will classify the AI and BD technologies application identified in the five "macro" areas (Task 1.3). (e.g. RD&I, process control, SC management, predictive maintenance and product customisation and traceability). A first mapping of AI and BD technologies and an industrial processes analysis per sector will be performed in WP1. A plan update for the subsequent WP of the project will be outlined, focusing on the AI and BD technologies application in the five "macro" areas. An extended and complementary literature review will be done to assure that the following areas are adequately assessed, while any other area not identified until this moment could also be taken into consideration. This classification will be considered during all the following work packages.

1.3 Concept and methodology; quality of the measures

1.3.1 Concept and approach

AI-CUBE concept

The AI-CUBE concept – as visualised more in detail in Figure 3 above – is based on a **tri-axial mapping of AI and BD technologies** in the different process industry sectors and macro-application-process areas. It allows to map and visualise the **STATUS** of AI & BD use and penetration per SPIRE sector (cement, ceramics, chemicals, engineering, minerals and ores, non-ferrous metals, steel and water) and per macro-application- process areas (e.g. research and innovation management and planning, process control, supply chain management, predictive maintenance, product customization). These application areas and related AI & BD technologies, will be pre-defined based on literature research, including research on previously funded projects, but cyclically updated during the project, when most promising business cases to be defined through a bottom-up engagement process with industry stakeholders, become clear. As confirmed by research[9] 57% of companies expect AI to have an impact on 'business areas' that are entirely unknown to us today, indicating that value drivers, pre-defined macro-application areas and future AI & BD business cases, are currently merely indicative.



The 'cube concept' allows to set the stage for the development and integration of four sets of key results which are key to **drive the opportunity assessment and take-up of AI and BD by the European process industries**:

1. **MLA:** A novel AI and BD Maturity Assessment Level model based on five activities: inception, elaboration, construction, deployment and maintenance.
2. **The AI-CUBE tool**: to be used as mapping tool by individual industries or the sector to assess their state of advancement in AI and BD, and needs for further development
3. **A set of inspirational AI and BD Business Cases in the process industries and a AI-CUBE Business Model Game**: to be used by industries or the sector, to stimulate reasoning about how AI and BD can change their Business Models and Value Chains for the future, and hence what actions to take
4. **Roadmaps and Infographics**: A set of several roadmaps and easy to consult Infographics, that outline the path towards:
    a. transferring good AI & BD practices form one SPIRE sector to another
    b. further developing and taking-up technology solutions towards AI & BD Business Cases of the future for the different SPIRE process industry sectors.

[9] EY Report "Artificial Intelligence in Europe. Outlook for 2019 and beyond

The AI-CUBE Approach

The methodological AI-CUBE approach (see Figure 4), uses the "cube-concept", to assess and visualise the **AI and BD STATUS** (3.1.) based on literature research and stock-taking with SPIRE industries stakeholders (1.1. and 2.1.). Based on the initial STATUS assessment in the **first phase** of the project, the definition of future AI and BD business cases in the **second phase**, resulting from the **MAMCA stakeholder engagement process**, as well as a detailed gap-analysis, in the **third phase** a set of actions to realise the targeted AI and BD strategies and roadmaps will be developed (4.3.).

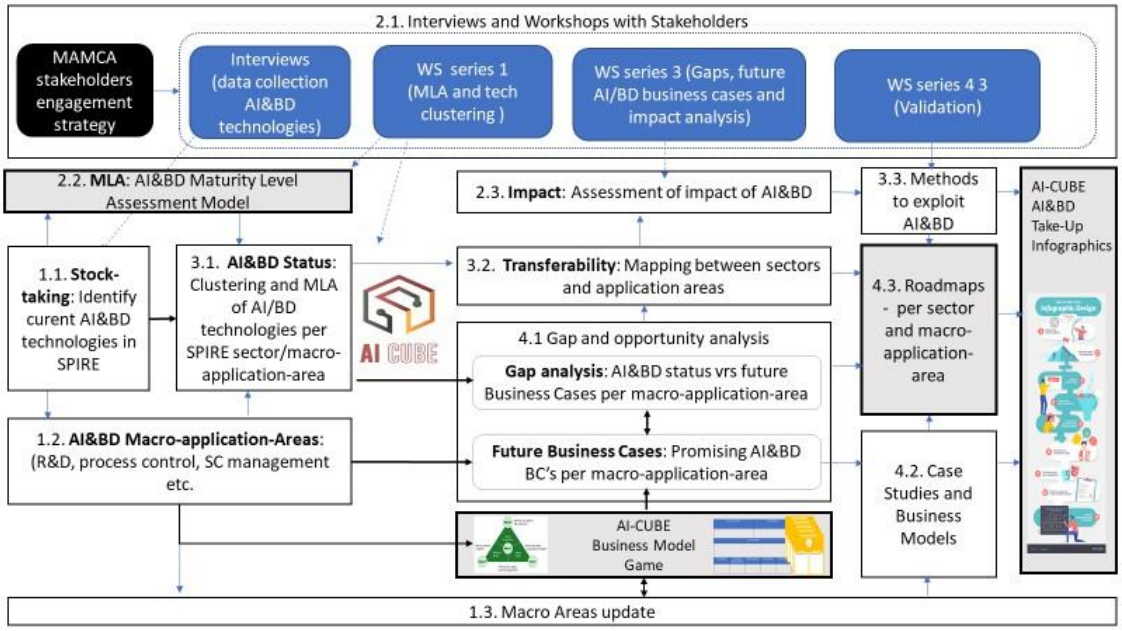


Figure 4. The AI-CUBE Approach

Once the **AI and BD STATUS** has been established, a **Maturity Level Assessment (MLA)** method and standards will be developed (2.2.) by the AI-CUBE team, and validated with the stakeholders' group through a **series of interviews and Workshops** (2.1.) following a Multi-Actor Multi-Criteria Analysis methodology. **This methodology** (see 1.3.3.1) will be used to obtain a widely supported and consensus-based action plan for the industrial consultation including a broad stakeholder community representing the main industry actors throughout all the SPIRE sectors. This stepwise and scientific approach allows AI-CUBE to involve a large group of stakeholders and interested actors in the process. The project consortium has strong connections with SPIRE industries and will use its support and experience to integrate all the sectors and reach all the involved actors by engaging strong and committed stakeholders. Several large process industries have already expressed their support to the AI-CUBE project (see letters of Support) and will be engaged in the process. The MultiActor Multi-Criteria Analysis (MAMCA) approach will be used to define the business areas, business cases and models, related organisational core processes and related AI &BD technologies, in relation to where the largest impact of AI and BD technologies is expected by the stakeholder communities, as well as gap analysis and validation of development requirements.

Based on an initial stock-taking and status assessment (the "CUBE"), together with industry stakeholders, the most promising **AI and BD Business Cases of the future** (4.1.) will be developed, taking into account both industrial competitiveness and sustainability objectives (SPIRE 2050 and Green Deal). A **Business Model Innovation Game**, developed and validated within the previous INSPIRE CSA, will be adapted for AI-CUBE to engage the industry stakeholders through a **"serious game approach",** to define the impact of AI and BD on their process industries. Using the CUBE, promising business cases and models will be mapped against **STATUS** (3.1) and a **gap analysis** will be carried out (4.1.). This will allow for the identification of AI and BD development areas, as well as those solutions for which good practices already exist in other SPIRE process industries, which can be adapted and **transferred** (3.2.). **Methods to transfer and better exploit AI and BD** will be defined (3.3.), and the **impact** (3.3.) of these potential developments will be assessed together with stakeholders through a series of Workshops with stakeholders (2.1.). The promising AI and BD Business Cases for the future, will be further analysed through **case-studies and business models** (4.2.) will be discussed with the stakeholder community, as input for the final results: the development of **individual roadmaps for each**

SPIRE process sector to progress towards these targeted AI and BD business cases. Finally, to foster dissemination and take-up of results, the AI-CUBE partners will design easy to use and digest **AI and BD Infographics**, to help realise one of the key final challenges of CSA type projects, to transfer the acquired knowledge and tools towards the wider (process industry) stakeholder community.

The AI-CUBE roadmaps will investigate:

How to identify the major business problems/opportunities and develop AI business strategies, based on the long-term strategic issue faced by the process industry in Europe? What type of AI to deploy, given the uncertainty regarding which technology will become widespread?

How to make use of the lessons learned from the implementation of AI solutions so far (most likely in technology-focused firms) in process industry? What would be the major/particular challenges that the process industry would face in deploying AI? AI-CUBE will analyse success stories and failures in similar settings and industrial applications (e.g., targeted marketing, dynamic pricing, manufacturing automation) to answer the question: How to manage AI related risks?

What kind of new technologies should be coupled with AI in sectors with heavy assets and long production cycle times, such as some sectors in process industry? The purpose is to identify the potential of such technologies (e.g., machine and deep learning, image recognition) in facilitating the implementation of the recommendations from AI solutions for achieving meaningful impact on different value chain functions (i.e., design, innovation, sourcing, making, storage, delivery, marketing, sales, use, after-sales, recycling/remanufacturing). This would help decision makers in process industry to take full advantage of AI, to close the "AI gap" and not to fall behind technology-focused industries.

What are the available methods/tools/processes to acquire relevant data? How can the companies in the process industry establish processes to identify the type of information necessary to improve financial/operational/environmental performance and how AI can contribute to the collection and analysis of such data/information? How to increase the involvement of data science in the decision-making process at higher levels?

How to scale deployment of AI technologies across industries/locations? What kind of AI ecosystems need to be established (e.g., IoT infrastructure, IT expertise, top AI talent)? How would this benefit not only single firms, but also value chains or industrial parks (e.g., industrial symbiosis, supply chain coordination)? How to guarantee that the process industry workforce is adequately trained in applying and benefiting from the use of data-driven technologies, such as AI and BD? How to recruit employees with the proper digital skill set?

How to determine the tasks/decisions that need to be automated? What would be the process governing the human-machine interaction (e.g., how would the recommendations from the AI solutions be handled, under what conditions would the human decision maker accept or reject the decision proposed by the AI solution)?

What would be the impact of the deployment of such AI technologies and BD on different stakeholders in a value chain? Would some players (e.g., freight forwarder or a distributor in a supply chain, humans in individual firms) run the risk of being disintermediated because of AI solutions? Do the AI solutions have the potential to create all-win solutions?

1.3.2 State of the art

1.3.2.1 State of art: Data driven technologies

In recent years, digitalization has become more and more important, so that it has a high significance in the manufacturing and planning processes of the process industry.

Europe is indeed the third country at world level for what concerns AI players and investments (see fig. 5) with a growing shift of investments both from the public and private sector.

In 5 years from now (see fig. 6), the market is expected to grow almost ten times compared to the market revenues in 5 years and, according to a study by IDG Research Services[10], up to 25% of companies would like to deal with the topic of artificial intelligence and machine learning in the coming years, so that it can be said that these topics already have a high influence in the economy today and their importance for future economic success is very present.

[10] Reder B (2018) Studie Machine Learning/Deep Learning.

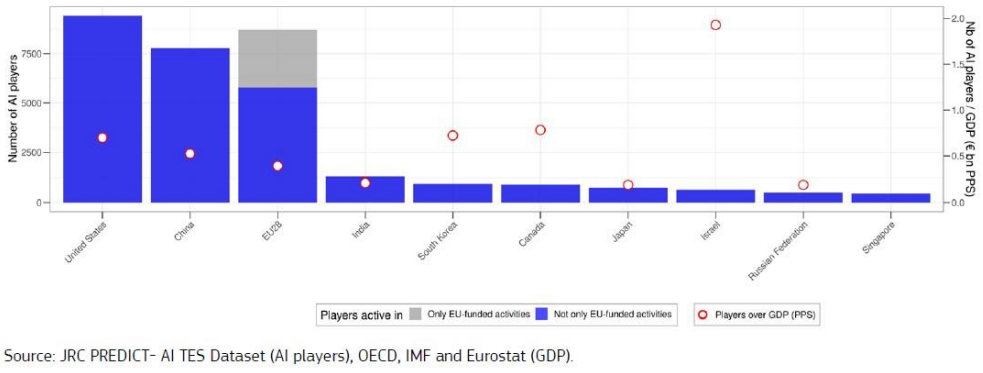


Figure 5. Top 10 world geographic areas by n. of AI players: Absolute&Relative to GDP (€ bn), 2009-2018.

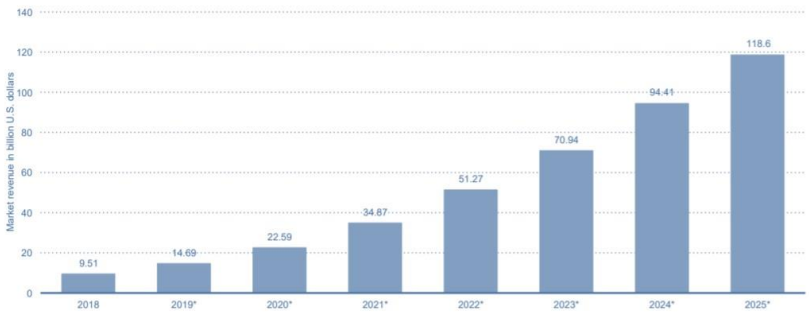


Figure 6. Expected growth of AI market revenue worldwide 2018-2025[11]

Borrowing the definition from the report "*Accenture Technology Vision 2018: Intelligence Enterprise Unleashed*", AI can be defined as "any form of machine learning". The ultimate goal is to obtain predictions and recommendations based on big data. "Deep learning" and "reinforcement learning" are two growing areas of machine learning, according to the same report (Accenture, 2018). These learning methods aim to recognize patterns in structured/unstructured data (e.g., images, text, videos, quantitative data) and make predictions and continuously improve (retrained prediction models) the performance by analysing new large data sets.

AI is considered a tool with great potential that will enable the firms increase profits in general. In fact, the results from a survey of over 3,000 business leaders in 2017, show that 85% of the respondents believes that AI will make their businesses remain competitive. Despite the promising solutions AI might provide, not all the businesses have fallen for the hype yet. Adoption rates are still quite low, around only 23% adopting some AI technology and a mere 5% having extensively deployed AI solutions and mostly in support functions such as IT and customer service[12].

Nevertheless, there are success stories in regards to the deployment of AI, not only in back-end IT solutions, but also core business functions (e.g., sourcing decisions, storage, delivery, design). In what follows, it is discussed **how the AI technologies have been used and the generate impact in different industries**:

- Use of AI powered automated machines by *Abundant Robotics*[14] to harvest apples, increasing the efficiency of the physically demanding apple picking process and avoiding issues with low labour supply,
- Use of AI (*Salesforce Einstein*) that prioritizes leads with a higher likelihood of converting to a sale, enhancing the sales process via enabling sales people to focus on the right leads increasing conversion rate in *U. S. Bank*,

[11] Statista, https://www.statista.com/study/68108/artificial-intelligence-in-italy

[12] Sam Ransbotham, David Kiron, Philipp Gerbert, and Martin Reeves, "Reshaping Business With Artificial Intelligence," MIT Sloan Management Review and The Boston Consulting Group, September 2017, p. 1. [14] https://www.abundantrobotics.com (accessed June 26, 2018).

- AI allowing *Stitch Fix* to analyse customer taste (data on colour, pattern) and purchasing behaviour (style), and consequently propose new personalized product designs that will match customer needs faster,
- Use of reinforcement learning by *Airbus/Autodesk* to make airplane partitions that are 45% lighter by applying patters for slime mould and mammal bones,
- Use of machine learning to provide customized products in less than 3 days, analysing customer photos and skin tones by *Shiseido/MatchCo*,
- Use of AI by *IBM's Watson* as a result of end-to-end visibility from data across systems, to spot obsolete parts identifying product changes and recommend procurement managers to order new/different parts leading to lower inventory levels and scrapping costs, as well as predicting which components have a risk of supply shortage to minimize disruption risks,
- Use of machine learning to improve maintenance and increase uptime of wind turbine machines by predicting near time failures from sensors placed on turbines by *MidAmerican Energy Company/Uptake*,
- Use of machine learning by *Big River Steel/Noodle.ai* to increase profitability by estimating prices of scrap and finished steel and projecting demand,
- Use of machine learning by *OhmConnect* to analyse house-hold level energy usage data, demographics and weather forecasts to optimize customer segmentation improving the efficacy of energy saving campaigns.

Artificial intelligence and machine learning have the potential to transform the management and structure of global value chains. New technologies in critical areas such as natural language processing, sensors, robotics, edge computing, machine and deep learning, and image recognition, provide significant opportunities to improve how products and services are designed, made, delivered, marketed, and used. Industry applications include targeted marketing, dynamic pricing, product design, manufacturing automation, and supply chain coordination.

Since data is so central to contemporary AI development, several data-related concerns are frequently raised during debates about AI. AI engineers spend as much time thinking about data as algorithms. They need lots of good quality data to perform effective ML, and even more to test the results. 'Data mining', is a field of computation focused on the automated identification of patterns and anomalies in datasets. The dataset could be anything from text posted on social media to precise measurements of underground geological formations, and the mining process could deploy ANNs, statistics and modelling to identify useful features. 'Big data' refers to datasets that are so large and complex – including content from different sources, in different formats, and with different degrees of authenticity and accuracy – that they cannot be stored or processed in the same way as smaller datasets. Figure 5 shows the different steps of implementing big data within a company. This reaches from ensuring the necessary infrastructure up to the involvement of the staff to prepare and enable a proper human-machine-interaction environment. Fig. 7 shows the early information activities to be conducted before actually implementing the first big data project.

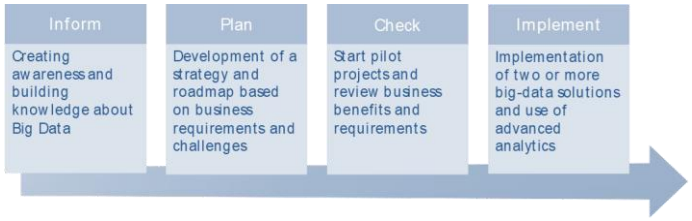


Figure 7. Implementation of Big Data in a company[13]

[13] IBM Global Business Services. Analytics: Big Data in der Praxis, Oxford 2012

Despite the potential of AI in helping companies create smart, resilient, and agile value chains, there are several reasons why no abundance of successful implementations of AI technologies can be found yet. Below, we present some of the **main barriers against the widespread deployment of AI**:

- Lack of AI expertise in areas such as data collection and preparation (disaggregation and formatting), data analysis, model training,
- Lack of IT infrastructure, talent, technologies and financial support for deployment of such technologies,
- Lack of AI solutions relevant for particular business needs (especially core functions),
- Existence of traditional systems with heavy assets and/or long production cycles, in which adopting AI solutions and integrating with the system is not as straightforward,
- Change management (especially when AI solutions introduce significant changes to the processes)
- Uncertainty regarding AI being a job creator or job destroyer, from a social welfare perspective for policy makers,
- Risk of AI-based decisions not always being well understood or justified, creating difficulties in decision making and accountability issues,
- Difficulties in implementing AI solutions in heavily regulated sectors (e.g., healthcare and financial technology), under strong data protection laws and privacy issues in relation to accessing and processing data,

Fig. 8 shows the adoption of AI in different sectors.

Apparently, some sectors are leading in the deployment of AI (e.g. high tech and telecom) whereas some, such as construction industry, have had limited use of AI to improve operations and manufacturing (see Fig. 6).

Our objective in this industry is to research what the role of AI could be for the SPIRE community in particular, and how the adoption rates could be improved for the process industry at large.

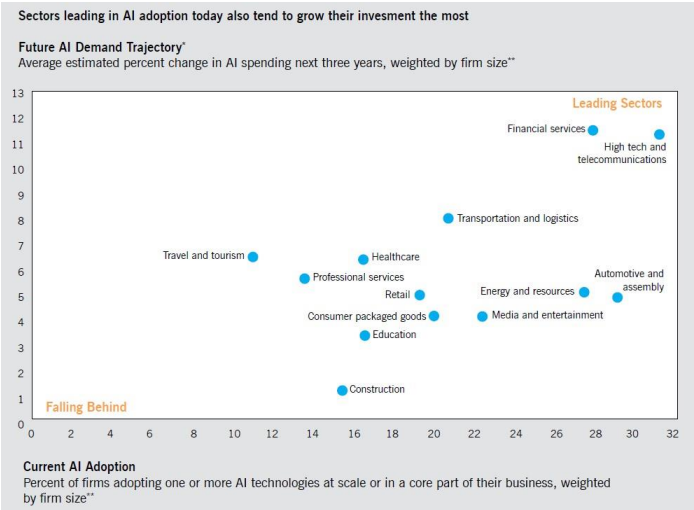


Figure 8. Sectors leading in AI adoption (source www.mckinsey.com)

Technological Gaps for AI

Early AI adopters are from sectors already investing at scale in related technologies, such as cloud services and big data. Those sectors are also at the frontier of digital assets and usage[14]. This suggests that the

[14] MGI, M. G. I., 2016. Digital Europe: Pushing the frontier, capturing the benefits.

technological gap will increase due to AI since there is limited evidence of sectors and firms catching up when it comes to digitization, as each new generation of tech builds on the previous one.

Independently of sectors, large companies tend to invest in AI faster at scale, also typical of digital adoption. Early adopters are not specializing in one type of technology. They go broader as they adopt multiple AI tools addressing a number of different use cases at the same time.

Companies investing at scale do it close to their core business. Early adopters that adopt at scale tend to be motivated as much by the upside growth potential of AI as they are by cutting costs. AI is not only about process automation, but is also used by companies as part of major product and service innovation. This has been the case for early adopters of digital technologies and suggests that AI-driven innovation will be a new source of productivity and may further expand the growing productivity and income gap between highperforming firms and those left behind[15].

Sectors leading in AI adoption today also tend to grow their investment the most. AI technology early adopters are industries with long histories of digital investment. Automotive and assembly was one of the first sectors that implemented advanced robotics at scale for manufacturing, and today is also using AI technologies to develop self-driving cars.
A major gap of AI is that the used algorithms have difficulties to understand the context of the data that they are handling. The patterns and correlations found by the algorithms often go unnoticed to human experts, so that decisions may even confound the engineers who created the algorithms[16].
For some AI systems missing creativity might also be a problem and steps need to be done for improving capability of natural language processing[17]. The high costs of implementation are also a key gap for AI.

The need for specialised workforce is growing, but statistical, technical and IT skills related to AI and data science are still rare when considering the entirety of computer science specialisations.[18,21] For these reasons, the lack of human experts with specialised knowledge need to be investigated also in process industry and new paths need to be defined in order to fill the skill gaps.

Implementation Challenges for AI

Important issues for AI are the data collection, access and processes therefore very much in relation with BD handling. AI algorithms work best when there are vast quantities of data, available in real-time. Therefore, a high data quality and availability and the implementation of the right algorithms and learning processes, are the biggest challenges for AI [19]. In many cases, process industries do not have enough data available to apply machine learning techniques in a suitable way. New technologies must be developed which are able to work with a significantly lower amount of data to produce similar results. Evolution from “Big Data techniques” to “Smart Data methodologies” combining process modelling and AI learning. Furthermore, even “Smart Data” are not available due to the difficulty of generating data points; for instance, when it requires building physical prototypes for testing, or complex simulations that may take days to run. In these cases, it is desirable to be able to start building models with very few data points, here called “Small Data”, and continue learning as new data become available. From a social point of view, the automation or “robotisation” is another implementation challenge not only for process industry but in general for societies and governments to jointly

[15] Zeebroeck, J. B. a. N. v., 2017. The right response to digital disruption. MIT Sloan Management Review.

[16] Dickson, B. (2017), “Artificial Intelligence has to deal with its transparency problems”, available at: https://thenextweb.com/artificial-intelligence/2017/04/23/artificial-intelligence-has-to-deal-with-its-transparency-problems/

[17] Vosloo, S. (2018), “Alexa, When Will Natural Language Processing Revolutionize ICT4D? Your Weekend Long Reads”, available at: https://www.ictworks.org/natural-language-processing-revolutionize-ict4d/#.WuHUzG4vz3g

[18] Smit, J., Kreutzer, S., Moeller, C. and Carlberg, M., *Industry 4.0*, European Parliament's Committee on Industry, Industry, Research and Energy (ITRE). Brussels: Policy departments A: Economic and Scientific Policy [21] https://www.worldmanufacturingforum.org/key-recommendations-2019

[19] Wuest, T., Weimer, D., Irgens, C. and Thoben, K.-D. (2016), “Machine learning in manufacturing: advantages, challenges, and applications”, *Production & Manufacturing Research*, Vol. 4 No. 1, pp. 23–45

develop mechanisms to adapt to these changes and some steps for understanding the ethical implication of this approach are already in place but need to be developed and customised for the process industry context[20].

Technology Gaps for BD

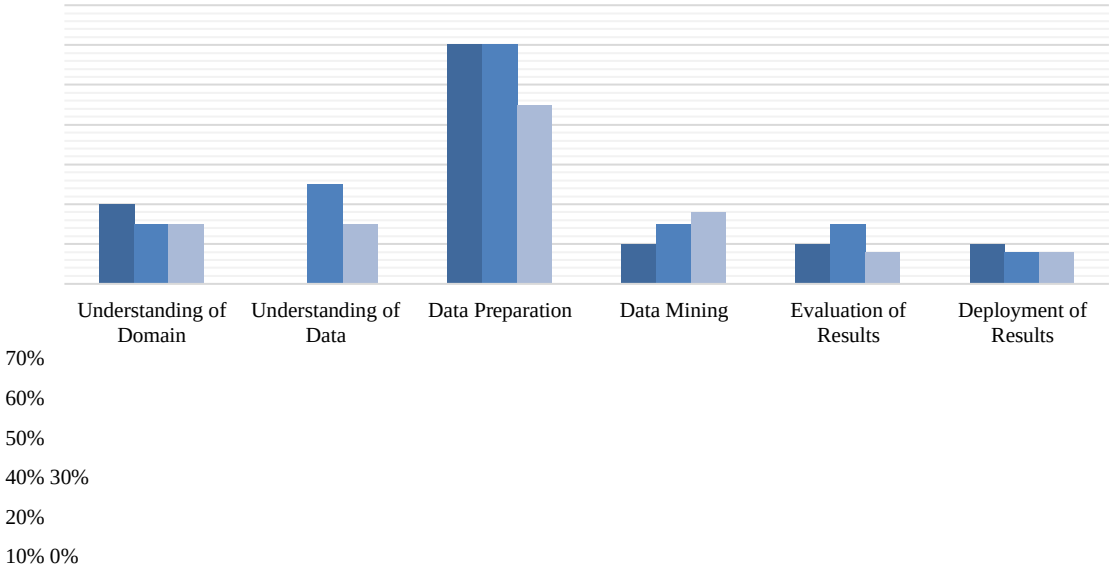
Another technology gap is the collection, storage and processing of the vast amounts of data itself. Terabytes of data will take large amount of time to get uploaded in cloud. Moreover, this data is rapidly changing rendering it hard to be uploaded in real-time. Most data storage technologies provide a different interface and abstraction as relational databases, which increases the complexity in adapting existing applications also in process industry. At the same time, the cloud's distributed nature is also problematic for big data analysis. Heterogeneous and incomplete data are big problems as the algorithms expect homogeneous data and cannot understand nuances. Therefore, it is not possible to devise absolutely fool proof, fully reliable, fault tolerant machines or software. When talking about BD, it is also important to consider the related problems of cybersecurity.

Implementation Challenges for BD

Regarding the technology gaps, the main implementation challenges are the reduction of the probability of failure to an acceptable level, and the creation of storage and processing capabilities for the vast amounts of data. Therefore, the definition of the IT architecture, the easy access to and use of data sources are decisive [207]. The testing of the data-intensive algorithms and systems will also be a big challenge[21] .
As the application of data science increases the complexity in data and computation governance, the control of data integrity, security and distribution, as well as the development of standards are other important challenges for implementation.

Data/information sources and data/information collection methods and practices – Big data to support AI

All AI methods are highly dependent on the necessary input information. This information is in many cases a large amount of data but could also be several forms of knowledge like rules or semantic descriptions of knowledge domains. The ways and means to collect all this information and to store it in a suitable way are of high importance for the success of all AI technologies[22].



[20] Wright, Scott A., and Ainslie E. Schultz. "The rising tide of artificial intelligence and business automation: Developing an ethical framework." *Business Horizons* 61.6 (2018): 823-832.

[21] Big Data Value Association (BDVA) (2018), "Big data challenges in smart manufacturing Discussion Paper on Big Data challenges for BDVA and EFFRA Research & Innovation roadmaps alignment". Available at: http://bdva.eu .

[22] Source: Kurgan LA, Musilek P (2006) A survey of Knowledge Discovery and Data Mining process models. The Knowledge Engineering Review 21:1–24.

1.3.2.2 State of art: Data-driven technologies in the process industry

In its position paper regarding the view of AI in the process industry[23], SPIRE PPP identified the potential of AI technologies to support the Process Industry. Nevertheless, it also highlighted the fact that process industry was lagging behind in terms of AI implementation.

While different SPIRE projects funding by Horizon 2020 have focused on areas such as advanced process control, digital platforms or industrial symbiosis, next steps in R&D for process industry should tackle AI applied to Process Industry.

Indeed, data driven technologies have the potential to support process industry companies to adapt to disruptive transformations in different industrial processes (product development, plant and supply chain operation).

The following **general AI application areas in the process industry** can be identified up to now:

- Modelling of static and dynamic systems while coping with uncertainty
- Detection of causality and correlation (e.g., of quality defects)
- Realisation of smart sensors (e.g., in the field of vision, as inputs for higher level systems like PLCcontrollers)
- Intelligent plant and process monitoring
- Online process control and optimisation
- Supervisory control such as through-process product quality control
- Control of industrial robots ensuring their capability for collaborative work together with human operators
- Decision Support Systems for repetitive tasks which require higher level cognitive operations (e.g. situation awareness) in combination with complex data processing tasks •
  Applications in predictive maintenance.

All these applications are oriented to the technical processes and their control and optimisation. Furthermore, there are applications which do not differ from other industries such as manufacturing industry and which deal, for example, with process mining of business processes or machine learning techniques to analyse future behaviour of customers based on intelligent analysis of information sources like the World Wide Web. However, the required processing of raw data, or even of information, refers to specific ontologies and issues.

[23] Artificial Intelligence in the Process Industry – A view from the SPIRE PPP (2019)

The above-mentioned application areas are driven by AI technologies which do not differ from AI applications in other technological fields, such as other industrial areas or telecommunication or social networks, for example. Nevertheless, there are significant differences, not in the technologies themselves, but more in the preparation of the necessary input information like data or knowledge, and in the way to implement them in a complex environment for online purposes.

The **most commonly used AI technologies** are (see fig. 10):

- the different forms of **machine learning** such as supervised (e.g. deep learning neural networks or random forest of decision trees or Support Vector Machines, etc.) and unsupervised (e.g. clustering techniques like SOM/DBSCAN or dimension reduction techniques like AutoEncoder)
- **reinforcement learning** (e.g. Deep Q Network or Q-Learning, etc.)
- all the techniques of **machine-based reasoning**, like Expert System Technologies (rule-based knowledge description in combination with inference mechanisms) or **other knowledge driven techniques** like Ontologies (semantic knowledge descriptions) also in combination with inference machines
- an emerging field is the combination of AI methods with other technology like **Multi-Agent-Systems**, which can help to solve different kinds of optimisation problems in complex environments because they can be excellently adapted to the needs of the highly distributed systems used in industrial production.

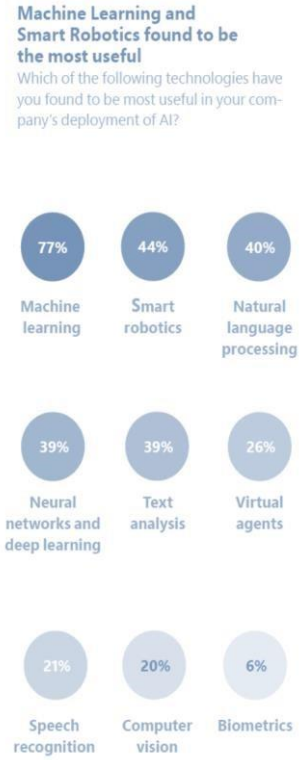


Figure 10. Most commonly used AI

For process industries, it is important is to be able to combine different AI methods with more classical ways of modelling (like 1st principal or statistical models) or other IT techniques.

The European Commission (EC) has put in place several initiatives on the AI landscape and a deep understanding of how it can impact on socio-economic system, offering useful insights for possible policy initiatives. The analysis identified six non-mutually exclusive thematic key areas within the AI technological domain, used to describe the strengths of economic players and geographical areas in AI subdomains. These thematic areas are: (i) Natural Language Processing (NLP); (ii) Computer Vision, (iii) Machine Learning (ML), (iv) Robotics and Automation, (v) Connected and Automated Vehicles (CAVs), (vi) AI Applications and Services.

The analysis implemented in the EC AI watch project[24] is conceived as an analytical framework and replicable methodology to analyse and describe the dynamics of specific ecosystems, by exploiting different types of factual data including non-official heterogeneous sources. This approach helps mapping AI technologies, monitoring and benchmarking its main players, their locations and R&D activities. One of the results is the creation of the thematic profile per each geographical area considering the distribution of the activities related to the six areas between 2009 and 2018. The analysis maps the ecosystem of AI in Europe compared with major competitors worldwide. This can be a starting point for the technology assessment in the AI-CUBE project that has to consider the activities and applications of AI and BD in the SPIRE sectors: cement, ceramics, chemicals, engineering, non-ferrous metals, minerals, steel, water.

A lot has been told on the role of AI in manufacturing, logistics, and other value chain phases, and about organizations' future plans to deploy the technology. Even so, current information on the actual AI technology

[24] The AI Techno-Economic Segment Analysis. https://publications.jrc.ec.europa.eu/repository/bitstream/JRC118071/jrc118071_the_ai_techno-economic_segment_analysis_1.pdf

deployment in the industry is limited and survey data show that this deployment is low, particularly in SME and industries where information technology is not core to the business[25]. AI adoption is greatest in sectors that are already strong digital adopters. At the same time, AI and BD technologies are being planned, piloted and deployed increasingly in all sectors, including the process and manufacturing sectors (see fig. 11 with an assessment of the status of AI maturity, as analysed by EY in a recent Report "Artificial Intelligence in Europe Outlook for 2019 and beyond") 61% of companies is still in the planning or piloting phase.

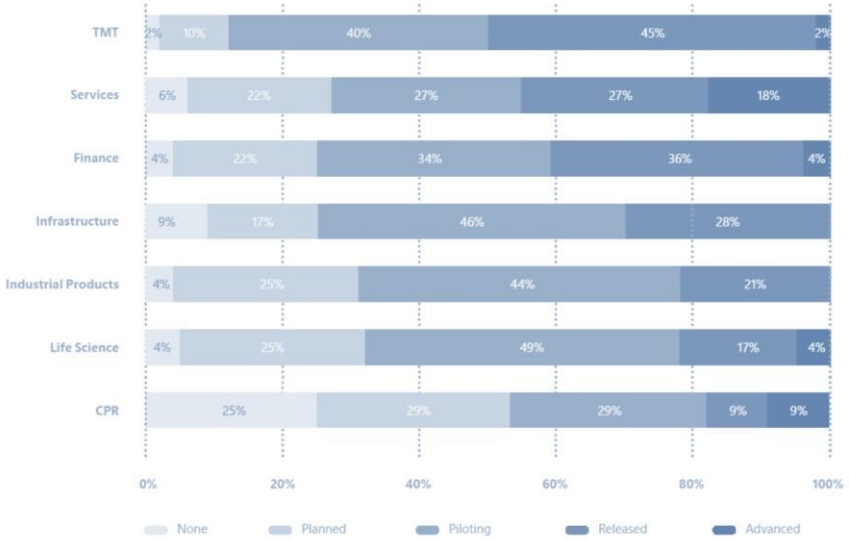


Figure 11. Assessment of the status of AI maturity

Fig. 12 provides a summary of how AI technologies can create value by driving process efficiency, process enhancement, and/or product process innovation:

SUMMARY OF VALUE DRIVERS				
Design	**Source/Make**	**Deliver/Store**	**Sell**	**Use**
Optimize product design basd on predicted customer behavior	Increase efficiency	Reduce delivery times	Analyze store foot traffic	Increase efficiency
Optimize product design based on desired criteria	Increase machine uptime	Improve fuel efficiency	Improve conversion	Automate customer service
Iterate product design faster	Minimize input price risk	Increase warehouse efficiency	Improve conversion	Anticipate maintenance
Personalize product	Reduce delivery times	Reduce inventory	Personalize marketing	
	Optimize input purchase timing	Avoid weather-related delays		

Process Efficiency Process Enhancement Product or Service Innovation

Figure 12. Summary of value drivers: AI creates value by driving process efficiency, process enhancement, or product/process innovation[26].

[25] Hau L. Lee, H. M. S. R. L. B., 2018. Value Chain Innovation: The Promise of AI. Value Chain Innovation Initiative 2018.

[26] Hau L. Lee, H. M. S. R. L. B., 2018. Value Chain Innovation: The Promise of AI. Value Chain Innovation Initiative 2018.

1.3.3 Methodologies

AI-CUBE will explore data driven technologies, AI and BD applications and maturity levels in process industry sectors, based on **a combination of different methodologies** that will lead to key results including the drafting and validation of a roadmap and guidelines to drive the implementation of AI and BD technologies across mentioned sectors.

1.3.3.1 Data-driven innovation intelligence and stakeholder analysis

The starting point of the exploration activities within AI-CUBE will be the systematic analysis of technological trends and stakeholder landscape across the eight process industry sectors of SPIRE. Partners will conduct indepth **systematic literature reviews** to systematize extant knowledge in the fields and identify potential technological trends to be explored. The output of this first search will be an overview of the main research and innovation trends in the field of data-driven technologies in the process industry. This will be compiled using not only the review and analysis of scientific and sectorial literature, but also carrying out a data-driven analytical assessment on AI and BD relevant projects funded by the EU, European Countries, the US and Canada, as well as patents granted in the last 5-10 years. Partners will make use of PNO's proprietary dataanalysis tool, WheesBee (see Fig. 13), featuring more than 500.000 publicly funded projects from various EU programmes (incl. FP7, H2020) as well as various national RD&I funding programmes from EU member states, the United States and Canada.

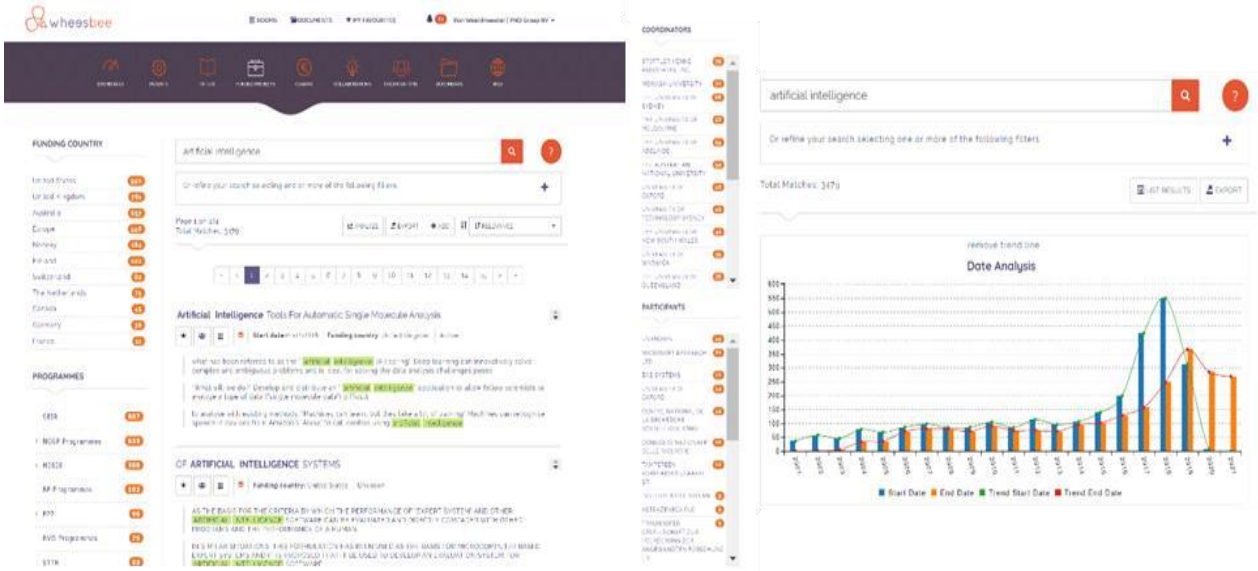


Figure 13. PNO's data-driven Innovation Intelligence tool with globally funded RD&I projects

The statistics will show how the number of funded projects and granted patents varies over time and in different countries. These analyses will be complemented by the exploration of the European process industry stakeholder scenario, in order to identify relevant actors and potential best practices and case studies (to be explored in T4.2) in the application of / R&D on AI and BD technologies. The goal is to shed light on what data-driven technologies are applied, in which industrial sector and/or process, by who and where; which are the drivers for and barriers to technological development and application, and how they can be boosted. These analyses will provide elements to be assessed in terms of levels of penetration and maturity, and the results will therefore allow populating the AI-CUBE framework presented above, in Fig. 3.

Additionally, AI-CUBE will encompass the following specific methodologies:

- Multi-Actor Multi-Criteria Analysis
- Maturity Level Assessment (MLA) Model
- ELSI (Ethical, Legal and Social Implications)

1.3.3.2 Multi-Actor Multi-Criteria Analysis

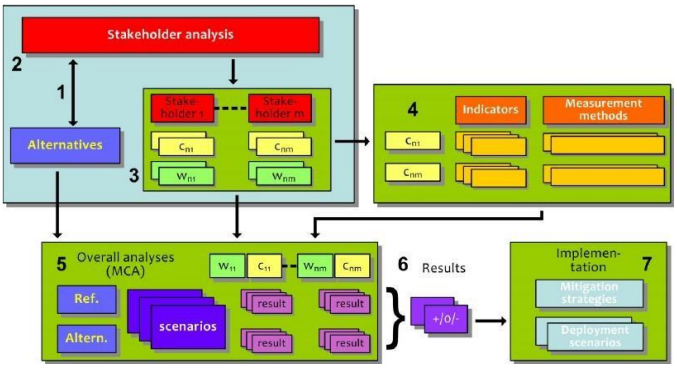


Figure 14. MAMCA methodology[1].

In order to obtain a widely supported and consensus–based action plan, the Multi-Actor Multi-Criteria Analysis (MAMCA) methodology[27] is used to consult a broad stakeholder community representing the main sectors of the SPIRE industry in Europe. This stepwise and scientifically sound approach allows the consortium of the project to involve a large group of stakeholders in the process of analysis if the AI and BD technologies implemented in the SPIRE industry.

MAMCA enables the evaluation of different alternatives (policy measures, business concepts, scenarios, technologies, etc.) by explicitly taking into account the objectives and framework conditions of the actors involved in the decision-making process. MAMCA thus develops a separate value tree for each stakeholder instead of just one value tree for all stakeholders (MCDA), which allows an individual view of the results for each stakeholder.

In order to get a more detailed understanding of the MAMCA method, the seven phases shown in fig. 14 are briefly described.

- Step 1: Definition of the problem and the identification of the alternatives
- Step 2: Stakeholder identification and in-depth understanding of each stakeholder group's objectives
- Step 3: Definition of criteria and their weights
  - *Criteria*: goals and objectives of the stakeholder
  - *Weights*: importance
- Step 4: Creation of indicators (evaluation criteria) ○ Step 5: Construction of an evaluation matrix, aggregating each alternative contribution to the objectives of all stakeholders
- Step 6: Sensitivity analysis
- Step 7: Implementation and feedback loop

In Step 1, AI-CUBE will define the specific problems around maturity of AI and BD in the process industry; after that, the project will customize the steps 2 to 7 to the specific stakeholders involved (process industry, ICT community) and the relevant criteria (aligned with the Maturity Level Assessment described below). There are other evaluation methods such as CBA (Cost/Benefit Analysis), CEA (Cost-Effectiveness Analysis) or EEA (Economic Effects Analysis); however, these methods put the economic effects in the focus, so that the MAMCA method results to follow a more holistic perspective.

1.3.3.3 Maturity Level Assessment (MLA) Model

The definition of the Maturity Level Assessment (MLA) Model is based on five activities: inception, elaboration, construction, deployment and maintenance[28]. The model can be used for three purposes[29]:

1. to assess the as-is situation of the organization/process;

[27] Macharis, Cathy & Witte, Astrid & Ampe, Jeroen. (2009). The Multi-Actor, Multi-Criteria Analysis Methodology (MAMCA) for the Evaluation of Transport Projects: Theory and Practice. Journal of Advanced Transportation. 43. 183 - 202. 10.1002/atr.5670430206.

[28] Garcia-Mireles, G.A., Moraga, M.A., Garcia, F.: Development of maturity models: a systematic literature review. In: 16th International Conference on Evaluation Assessment in Software Engineering (EASE 2012), pp. 279–283 (2012)

[29] De Carolis, A., Macchi, M., Negri, E., & Terzi, S. (2017, September). A maturity model for assessing the digital readiness of manufacturing companies. In *IFIP International Conference on Advances in Production Management Systems* (pp. 13-20). Springer, Cham.

2. to approach maturity improvement in order to positively affect business value;
3. to enable benchmarking across companies; in particular, a model of this nature would be able to compare similar practices across organizations.

In the literature, there are some MLA models to evaluate the overall digital maturity level of a company, where it is possible to obtain the digitalisation level of a company with a benchmark comparison to the rest of the population analysed. These are systems that gain importance as far as they manage to involve larger number of cases. For example, the assessment of the DREAMY model (Digital REadiness Assessment MaturitY model), developed by the Industry 4.0 observatory of Politecnico di Milano, is based on business processes grouped in strategic areas for the digital transformation: (1) Design and Engineering; (2) Production Management; (3) Quality Management; (4) Maintenance Management; and (5) Logistics Management. Each of these processes is evaluated based on different analysis dimension: Process, Monitoring and Control, Technology and Organization. The model has 5 levels of maturity named: initial, managed, defined, integrated and interoperable, and digital oriented.

Another important model is the Industry 4.0 Maturity Model developed by Schumacher et al.[33], including a total of 62 maturity items which are grouped into nine company dimensions: strategy, leadership, customers, products, operations, culture, people, governance, technology. The definition of maturity level is based on the mathematical relationship between the considered factors and the related weights given by the community under consideration.
To go beyond these maturity level models for digitalization, there are already some specific initiatives related to AI and BD: McKinsey has developed an AI readiness index to compare the adoption and diffusion of AI at country level (considering in EU and USA). The system is based on the measurement of an AI Readiness Index[34] based on: the number of AI startups per capita, automation potential of job activities, digital maturity, the availability of scientists and engineers, ICT business model creation, R&D expenditure, ICT connectedness.

Capgemini has developed a tool to map the readiness and the performance of the countries in the AI field. The AI readiness is the extent to which a country and its institutions & businesses have the ability to reap the benefits of AI; the dimensions considered are: (1) institutional environment, (2) IT maturity and (3) IT Skill Advancement. Five components have been defined as the foundation of AI performance: (1) Personnel, (2) Monetary Impact, (3) Competitiveness, (4) Research & Education and (5) Technology. [1]

All these MLA models will be analysed in details and a new **framework specifically for AI and BD penetration on the process industries** will be developed to: (i) provide an impartial and complete evaluation system updated with the state-of-the-art; (ii) understand the industrial needs in terms of technologies (iii) identify a cross-sectorial comparative system to define maturity level and furthermore suggest new actions for improvement. The framework will take into consideration the technological maturity of a company weighting the level of usage with the level of diffusion in the company culture and with the awareness of the workers. A special session of the model will consider also criteria to evaluate maturity towards ELSI (Ethical, Legal, Social Issues). This framework will be developed in WP2, and used and applied in WP3.
Within the cube approach, some dimensions to be taken into consideration to evaluate the readiness to AI of a company, and then of a sector, are related to: (1) Strategy definition, (2) Smart factory, (3) Smart operations, (4) Smart products, (5) Data-driven services, (6) Employees, and (7) Supply Chain.

[33] Schumacher, A., Erol, S., & Sihn, W. (2016). A maturity model for assessing Industry 4.0 readiness and maturity of manufacturing enterprises. *Procedia Cirp*, *52*, 161-166
[34] McKinsey Global Institute (2019). Notes from the AI frontier: Tackling Europe's gap in digital and AI.
https://www.mckinsey.com/~/media/mckinsey/featured%20insights/artificial%20intelligence/tackling%20europes%20gap%20in%20digital%20and%20ai/mgi-tackling-europes-gap-in-digital-and-ai-feb-2019-vf.ashx

[1] Capgemini Consulting (2018). Artificial Intelligence Benchmark. https://www.capgemini.com/wp-content/uploads/2018/07/AIReadiness-Benchmark-POV.pdf

1.3.3.4 AI and BD ELSI (Ethical, Legal and Social Implications)

Artificial intelligence and big data technologies will significantly impact society, values and human rights[30]. What will change for the better, and what will change for the worse? Which values could change and who will be most affected?

The EDPS Opinion reflected in the Ethics Advisory Group Report 2018, identifies several technological trends that require a rethinking of the relation between technology and human values, thus calling for the formulation of a 'new digital ethics': big data generated from a variety of sources, from public administrations and private companies, social networks and other online platforms, the internet of things and networked sensors, cloud computing, and artificial intelligence, in particular machine learning. These digital technologies require what the Opinion calls a 'big data protection ecosystem': an interactive and accountable assemblage of 'futureoriented regulation', 'accountable controllers', 'privacy-conscious engineering', and 'empowered individuals'.

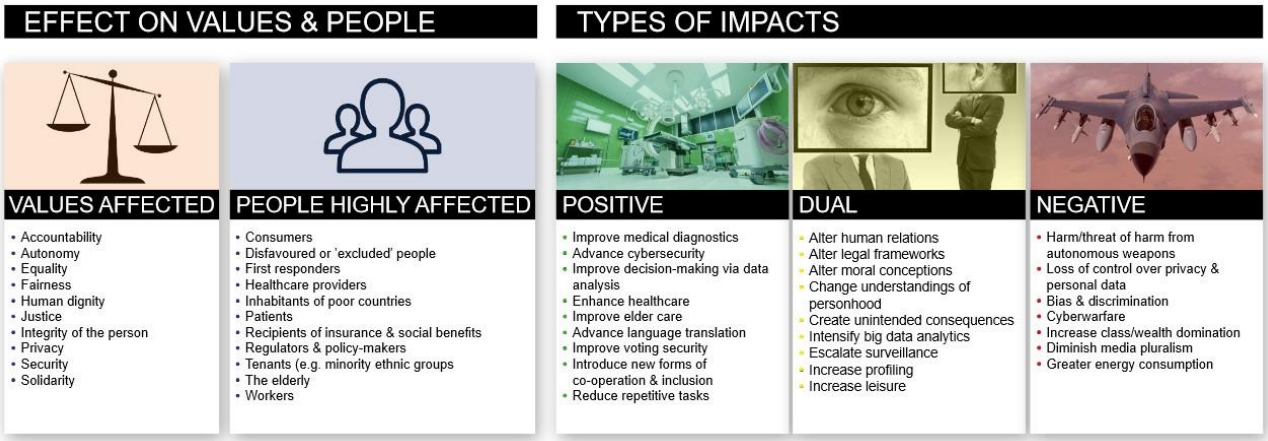


Figure 15. ELSI impacts[36]

The right to data protection is key to regulating a digitised industry. In light of recent technological developments, such a right appears insufficient to understand and address all the ethical challenges brought about by digital technologies.

AI-CUBE will revise the ethical, legal and social implications of AI and BD developments, considering the values encoded in algorithms, the need to evaluate outcomes, and issues of bias and transferability, data ownership, confidentiality and consent, and legal, moral and professional responsibility. The potential effects for clients and different actors in the supply chain, including confidentiality.

1.3.4 Relation to previous initiatives

The AI-CUBE project will exploit the strong synergies existing with the A.SPIRE project INSPIRE (H2020, Grant Agreement n°723748), focused on the development of new business models for flexible and sustainable manufacturing value chains, along with the definition of guidelines to measure these business models performance in different scenarios. INSPIRE greatly involved the stakeholders of the process industry and discrete manufacturing both in the design of the innovative business model and their performance assessment in different scenarios. This way, European process industry and discrete manufacturing sector would be

[30] Whittlestone, J. Nyrup, R. Alexandrova, A. Dihal, K. Cave, S. (2019) Ethical and societal implications of algorithms, data, and artificial intelligence: a roadmap for research. London: Nuffield Foundation. [3636] https://www.sienna-project.eu/robotics/elsi/

enabled in flexibly adapting to the proposed paradigms and catching value from highly dynamic future trends. AI and BD are among these trends and, building on the results and methodologies developed by INSPIRE, the AI-CUBE project will conduct an effective and comprehensive assessment of the state of maturity and penetration of AI and BD within the different levels of process industry organizations. Moreover, AI-CUBE will also involve stakeholders into the consolidation of this assessment and, finally, the identification and validation of guidelines for a cross-sectoral implementation and diffusion of AI and BD technologies. The AICUBE project aims at creating synergies with and leverage on the INSPIRE European Process Industry Forum, in order to actively involve stakeholder from the more than 100 European Industrial Parks that INSPIRE partners already approached individually. The AI-CUBE project plans to tap into this existing platform to reach effective collaboration with interested stakeholders of process industry sectors, but also dissemination and communication targets.

Concerning the macro-area Supply chain (SC), the project will benefit of the results of NEXT-NET project (H2020, Grant Agreement No. 768884), where a set of SC strategies have been identified and related research and innovation priorities as well, where some of them are strictly related to AI&BD. The paths identified in NEXT-NET will be further developed and specified specifically for the process industry.

AI-CUBE will also build upon the results of the following projects:

Table 1. Projects related to data driven technologies and process industry already identified by AI-CUBE.

Acronym	Project name	SPIRE sectors	Description	R&D	Production process	Supply chain	Maintenance	Customisation
NanoSolveIT	Innovative Nanoinformatics models and tools: towards a Solid, verified and Integrated Approach to Predictive (eco)Toxicology (NanoSolveIT)	*CHEMICALS*	Ground-breaking in silico Integrated Approach to Testing and Assessment (IATA) for the environmental health and safety of Nanomaterials (NM). Decision support system packaged as a standalone open software and a Cloud platform	**X**				
i-TRIBOMAT	Intelligent Open Test Bed for Materials Tribological Characterisation Services	*ENERGY* *STEEL* *CHEMICALS*	IT-platform for materials and tribological data harmonisation, management, analytics, sharing and mining. Lab-to-field upscaling tools combining testing with computation	**X**				
ELECTRO-POM	From discovery to scale up of cluster based electrolytes for Ultra-high energy storage flow batteries	*CHEMICALS*	AI driven discovery system of the ERC Advanced Grant SMART-POM to discoveR new metal oxide molecular polyoxometalate (POM) clusters	**X**				
SMART-POM	Artificial-Intelligence Driven Discovery and Synthesis of Polyoxometalate Clusters	*CHEMICALS*	New platform for the preparation, understanding, and exploitation of precisely defined nano-molecules / polyoxometalates using automated intelligent feedback	**X**				
AITEKIN	Combination of AI techniques and software with advanced reactor equipment for efficient kinetics analysis in the chemical industry	*CHEMICAL*	Combination of AI techniques and software with advanced reactor equipment for efficient kinetics analysis in the chemical industry	**X**				
INEVITABLE	Optimization and performance improving in metal industry by digital technologies	STEEL NON FERROUS	High-level supervisory control systems for different production plants to enable autonomous operation of the processes based on embedded cognitive reasoning.		**X**			

MONSOON	MOdel based coNtrol framework for Site-wide OptmizatiON of dataintensive processes	STEEL	Multi-scale model based predictive controls on predictive monitoring of plotlines, targeting early detection of anomalies and potential optimization gains		X		X	
FUDIPO	Future Directions of Production Planning and Optimized Energy- and Process Industries	CHEMICALS WATER	Machine learning tools for diagnostics, data reconciliation, and decision support, production planning and process optimization including model-based control		X		X	
FACTLOG	Energy-aware Factory Analytics for Process Industries	STEEL	ML and analytical tools at different level to create a cognitive factory to understand the behaviour of complex process systems		X		X	

Acronym	Project name	SPIRE sectors	Description	R&D	Production process	Supply chain	Maintenance	Customisation
CoPro	Improved energy and resource efficiency by better coordination of production in the process industries	CHEMICAL	Methods and tools for process monitoring and optimal dynamic planning, scheduling and control of plants, industrial sites and clusters under dynamic market conditions		X	X		
COGNI PLANT	Cognitive platform to enhance 360º performance and sustainability of the European process industry	CHEMICAL STEEL	Advanced data analytics to extract valuable information from processes and their effect on production plant's overall performance		X			
RETROFEED	Implementation of a smart RETROfitting framework in the process industry towards its operation with variable, biobased and circular FEEDstock	CERAMIC CEMENT STEEL	Decision support system able to perform a diagnosis of the impact in the process of different retrofitting solutions so plant managers and operators can decide on the most suitable retrofitting action for their companies		X		X	
ProPAT	Robust and affordable process control technologies for improving standards and optimising industrial operations	CERAMIC STEEL MINERALS	Versatile global control platform for data acquisition, data processing & mining and User Interface to measure properties of process streams and products, accurately and in real-time (self-learning & predictive)		X			
COGNITWIN	cognitive plants through proactive self-learning hybrid digital twins	STEEL	Digital twin of the process used to develop models with cognitive capability for self-learning and predictive maintenance		X		X	
HyperCOG	Hyperconnected Architecture for High Cognitive Production Plants	CEMENT CHEMICAL STEEL	Hyper-connected network of digital nodes to catch outstanding streams of data in real-time with high computing capabilities		X	X		
SHAREBOX	Secure Management Platform for Shared Process Resources	CERAMIC	New analysis and optimisation tools for flexible energy use and material flow integration to optimising symbiosis		X	X		
DIY4U	Open Innovation Digital Platform and Fablabs for Collaborative Design and Production of personalised / customised FMCG	CHEMICALS	Web-based portals to allow users to design their own products from home with a broad range of performance attributes and then have these produced and delivered in a "designed for you" way			X		X
DATAMINE 4.0	Advanced Data Modeling and Analysis Applied to next Generation Industry 4.0 settings and the Internet of Things	ENG	Modelling and analysis of data coming from complex industrial and organizational applications to identify hidden casuistic relationships to calibrate production and organizational control systems			X		X
ROSSINI	RObot enhanced SenSing, INtelligence and actuation to Improve job quality in manufacturing	ALL	Enhanced robot sensing capabilities for the deployment of AI to optimise productivity and safety, and natively collaborative manipulation technologies		X			
TOMOCON	Smart tomographic sensors for advanced industrial process control	ALL	Fundamentals of integrating imaging sensors into industrial processes and functional feasibility demonstration	x	x			

AccuWater	High accuracy water leakage and apparent loss detection	WATER	Intelligent BD Analytics solutions to reduce Water Losses, or Non-Revenue Water (NRW) and for related applications				x	x
ChemChain	Blockchain Platform to Track Chemicals along the Value Chain	CHEMICALS	Open-source global blockchain platform to transfer information on hazardous chemicals along the value chain		x	x		
PreCoM	Predictive Cognitive Maintenance Decision Support System	ALL	Sensors with BD analytics to track machine-tool performance forp redictive cognitive maintenance decision-support system	x			x	
IoTwins	Distributed Digital Twins for industrial SMEs: a big-data platform	ALL	Testbeds for digital twins in manufacturing and facility management sectors, integrating data from APIs, historical, embedded sensors and open data		x		x	
SHERLOCK	Seamless and safe human - centred robotic applications for novel collaborative workplaces	ALL	Soft robotics collaborative Production Station. Novel human – centred interaction, collaboration and awareness (AI enabled cognition)	x	x			

1.3.5 Gender dimension

The consortium supports gender balance and integrates the gender dimension in the ELSI implications.

2. Impact

2.1 Expected impacts

The process industry is a key one for the European economy (contributing to the EU Jobs and Growth agenda with 6.3 million jobs directly created and further 19 million indirectly created in Europe)[31]. SPIRE PPP aims at boosting investments in Europe generating global competitiveness for EU Process Industries, better jobs and welfare for its citizens. In order to do so, SPIRE PPP envisions "Connecting the dots – investing for smart integration of process industries across Europe". AI and BD can play a key role in connecting the dots for Process Industry. AI-CUBE will accelerate the take-up of AI and BD in this industry, thus translating into a cascading effect that will create value across Europe and down to the end users.

Expected impacts from the Work Programme

AI-CUBE project will address the expected impact in the following way:

Expected impact	AI-CUBE impact
Better exploitation of AI potential for all the different sectors in the process industries, and strategies for developing AI applications, including the generation of data	AI-CUBE consortium will apply a MAMCA methodology to guarantee the involvement of wide stakeholder numbers who will contribute in the validation of the roadmap and guidelines resulting from the project activities. These stakeholders will also contribute to the definition and validation of concepts and constructs throughout AICUBE development. This deep stakeholder involvement aims at achieving a widely shared understanding of AI and BD technology potential in the process industry, but also detecting procedures and steps for those technologies improvement and effective application, along with the identification of the necessary ecosystem (e.g., technological coupling, infrastructures, expertise, etc.) capable of boosting data-driven technologies application, scaling and diffusion across industry sectors and value chains.

[31] SPIRE 2050 Vision https://www.spire2030.eu/sites/default/files/users/user85/Vision_Document_V6_Pages_Online_0.pdf

Identification of existing and future data requirements for the development of data driven technologies (eg. ontologies and data formats, how to handle data heterogeneity, data availability, quality and reliability, access and platforms, security and confidentiality)	The above-mentioned collaboration with stakeholder of the sectors, identified through a dedicated analysis, will guarantee the involvement of actors with experience and knowledge of data-driven technologies and of the requisites they need to present, i.e. specs that data need to meet in terms of formats, availability, quality and reliability, etc. AI-CUBE activities will identify the methods, tools and processes already available to acquire relevant data; the processes through which firms can identify the type of information necessary to improve financial, operational, and environmental performance; the potential contribution of AI to the collection and analysis of such data / information and its involvement in the decision-making processes.
Seamless collaboration of human operators with process control systems and plants: acceptance, identification of skills gaps and ethical, legal & social implications (ELSI)	AI-CUBE identifies existing and possible fields of machine-humaninteraction (HMI) along with interviews. Workshops with experts are used to point out on respective acceptance levels and necessities to develop specific skills and adaptation of technologies to allow further integration of HMI. AI-CUBE will dedicate Task 4.4 to the revision of the ethical, legal and social implications of AI and BD developments. This task will revise the required skills for a seamless collaboration of human operators and will analyse the current skill gaps in the industry to get ready for new implementation.
Implementation and further elaboration of the strategic research and innovation agenda announced in the EC Communication on Artificial Intelligence	AI-CUBE embraces the combination of several methodological approach, including systematic analyses of the extant literature, technological trends, and stakeholder landscape, along with the application of MAMCA, to widely involve sector stakeholders in the formulation of constructs and their validation, and ELSI, to detect potential AI and BD technologies impact on society, values and human rights. This combination of approach and perspectives considered will provide a thorough understanding of the factors driving AI and BD technologies application in the process industry, and the potential barriers hindering their development and diffusion. Furthermore, AICUBE activities will shed light on the wider socio-economic-cultural impact that data-driven technologies can have. All these elements together allow the drafting of comprehensive roadmap not only to foster further application of the technologies, but also to guide strategic decision making both at the research and policy level, on the elaboration of RD&I activities in Europe.

Expected impacts of data driven technologies on process industry

Process industries are working in a “brown field” environment (as opposed to “green field” applications which means in a new-build factory), which means that existing plant, control and IT-systems cannot easily be replaced completely by new solutions. Therefore, interfaces between new systems and existing equipment are necessary. Keywords in this field are SCADA, MES-MOM, ERP, ISA95/IEC62264, Data Warehouse, Data Lakes, Cloud Computing, Edge Computing, noSQL, and Streaming technologies, etc.

In combination with the data and information collection, the problems of the amount, quality and reliability of data must be mentioned. In this sense, the demands of process industries are extremely high because any kind of wrong decisions related to little or poor data could lead to unstable production processes with unforeseeable and potentially dramatic consequences. Appropriate solutions to handle situations of poor data quality in an industrial environment do not exist today and are one of the main research needs for the future.

Up to now, typically only case-specific solutions have been developed and implemented, while no systematic approach exists to develop solutions for different types of problem and roll out these solutions subsequently to all possible application examples. This is strongly connected to the project-based implementation of AI solutions, with the consequence that no systematic long-term maintenance, update and re-check of the deployed AI solutions after the end of the project is undertaken.

Potential barriers, obstacles or factors endangering the achievements of the expected impacts

Potential barriers, obstacles or factors	Measures to be undertaken
Human: lack of the skilled personnel needed to implement AI and BD	WP3-T3.2 will identify technological gaps and implementation challenges as well as possible skill gaps with regard to human-machine-interaction. In the roadmapping phase 3 (WP4-T4.3), results from the previous task will be translated into development needs and recommendations in terms of skills, data and RD&I requirements. WP4-T4.4 will identify the skills required to allow a seamless collaboration of human operators
Organisational: lack of a competitive business case, clear and with strategic vision	WP4 will define AI and BD driven Business Cases and Business Models for the process industries of the future. A mapping of the future Business Cases and Models against the current STATUS (WP3) will be carried out to establish gaps and development needs per SPIRE process industry sector and per macroapplication process areas.
Technological: unbiased data, lack of interoperability and cybersecurity	T3.2 will result in concrete application possibilities developed individually for each sector. The task also identifies possible barriers for the use of an AI methods & BD technologies in each industry sector along with the transfer of solutions between the sectors. T3.2 provides an overview of the major challenges faced in the introduction of AI and BD methods and identifies areas in which overcoming these challenges result in an increasing economic value.
Legal: lack of regulation in security and privacy	T4.4 is devoted to the revision of the ethical, legal and social implications of AI and BD developments, considering the values encoded in algorithms, the need to
	evaluate outcomes, and issues of bias and transferability, data ownership, confidentiality and consent, and legal, moral and professional responsibility. The potential effects for clients and different actors in the supply chain (human operators), including privacy.

2.2 Measures to maximise impact

a) Dissemination and exploitation of results

Dissemination, Communication and Exploitation (DCE) activities play a key role in the AI-CUBE project, as they contribute to foster impact, both during the duration of its EC funded lifecycle as well as beyond it, during its possible business oriented follow up. The aim of DCE activities is the maximization of awareness of the results achieved by the AI-CUBE project and the generation of a sustainable dissemination and exploitation platform based on the involvement of different audiences that have an interest in the output of the project. The aim of AI-CUBE is to provide an overview of the state of play in AI and BD applications throughout the European process industry and deliver an in-depth research and innovation agenda, including indication on the priorities on which solution providers (e.g. universities, software developers and the process industries themselves) should focus in the upcoming decennia. The agenda would also include policy recommendation, outlining how policies may (or may not) favour data-driven technology developments and take-up in the European process industries, going towards increased competitiveness and sustainability of an industry underpinning all economic sectors in Europe, as its key material and solutions providers.

Therefore, targeted audiences for the AI-CUBE project are:

1) The **eight SPIRE process industries sectors** themselves, through A.SPIRE as well as the INSPIRE European Process Industry Forum A.SPIRE association has been already contacted and they have agreed to being invited to the Steering Committee. Furthermore, feedback from A.SPIRE Board of Directors will be facilitated thanks to the appointment of Ms Oonagh MacNerney, CEO of IRIS, as honorary member representing SMEs. A relevant set of more than 100 industrial parks already identified in the A.SPIRE project INSPIRE targeting novel Business Models for the process industry, amongst others based on digitalisation and sustainability. This platform will enable the EU process industry to share

knowledge and expertise achieved through the AI-CUBE Project concerning: 1) the level of maturity and penetration, and 2) the strategic exploitation of synergies and similarities in AI and BD technology application; and 3) guidelines and new business models for a flexible implementation across different process industry.

2) The **Value Chain Partners of the European process industries**. Being the process industry a key material supplier in the Value Chain, it increasingly integrates and tunes its Supply Chain Planning with Value Chain partners such as the (discrete) Manufacturing industries (e.g. automotive, electronics, etc.) and Logistics. Digitalisation and data-driven technologies will play an increasingly important role in optimising, dynamizing and (re)configuring the Value Chains in an ever more modularised process and manufacturing industry. The AI-CUBE project will leverage on the recently concluded NEXT-NET project, to share knowledge of possible AI and BD applications that foster such collaborations, using the stakeholder community of VC partners from Manufacturing and Logistics.
3) **AI solution providers**: the AI-CUBE project will provide an overview of the white spots for AI in the process industry and, hence, whether and how they can represent an opportunity to develop, offer and exploit AI solutions. The project will reach out to this community with relevant information and recommendations, also through easy to read infographics and informative video.
4) **Policy Makers**: Policies such as GDPR and standardisation of data-formats that facilitate AI applications across multiple VC partners play an important role in fostering AI applications. Policy makers are hence a relevant audience for recommendations of the AI-CUBE project.

The AI-CUBE Project will identify the most common AI and BD technologies and define their level of maturity and penetration in different EU process industry. In so doing, it will also identify the drivers and barriers for the diffusion of these technologies, analysing the synergies to be leveraged for a cross-sectoral implementation of the technologies, and the challenges embedded in their development and application.
The PPs strongly believe that, in order to create synergies among different process industries, it is crucial to make concrete showcases and disseminate project outcomes and knowledge available to an increasingly expanding audience. This includes networking partners and stakeholders achieved through an iterative stakeholder expansion strategy, based on the INSPIRE project, a list including more than 100 European stakeholders that have been individually approached by the members of INSPIRE to be involved in project activities. The process will select form the INSPIRE list of 100 the industrial stakeholder, users, orchestrators, and solution providers, which are already applying AI and BD technologies to a minimum extent. The process will then iteratively test and expand towards a wider EU network of stakeholders, industrial clusters and parks, creating a 'platform' of actors collaborating with and interested in the AI-CUBE Project activities. The platform will be made available in open access to the maximum extent possible, in compliance with Intellectual Property Rights (IPR), company policies, and trade or industrial secrets. Not shareable results, i.e. covered by confidentiality agreements, will be adapted and made available to all targeted groups (i.e. 8 SPIRE process industry sectors, Value Chain partners of the European process industry, AI solution providers, and Policy Makers). The relevant project material will be made available through the AI-CUBE Project website and published through the project communication channels, including newsletters and other public materials. The coordinator, PNO, together with the PPs, will organize the dissemination so to obtain the maximum reach while controlling licensing and IPR issues related to the contents.

The dissemination and exploitation strategy consist of three separate dissemination expansion phases:

1. **Inward Oriented Exploitation and Dissemination:**
   a. Validation of the AI-CUBE business models, tools, knowledge base, and solutions in a contained set of stakeholders, identified through an ad hoc stakeholder analysis. In this phase, the stakeholder analysis will also be complemented with the MAMCA methodology in order to reach wider support and consensus to the activities of the project and consult a broad community of stakeholders representing the main sectors of the SPIRE industry in Europe.

b. The analysis will start from the members of the 100 European industrial parks included in the INSPIRE lists already mentioned and will further be refined by focusing on the stakeholders that are already applying AI and BD technologies to a minimum extent. This specific focus on AI and BD aspects will allow the definition of a subset of relevant process industries potentially interested in the activities of the AI-CUBE Project. From this initially limited number (e.g. 3) of process industry sectors, the Project will identify good practices and tools to further explore in a larger cross-sectoral European-wide network. Indeed, in the first phase of the AI-CUBE Project, a European-wide study will be conducted in the WP1, with 50-100 relevant stakeholders to collect relevant practices and design potential interviews and workshops to be developed in the WP2 (2.1)
c. In this phase, dissemination will focus on general communication about the AI-CUBE Project to the general public (e.g. through the website) in order to make knowledge, business solutions and practices available to facilitate the smaller initial group of stakeholders and its enhancement.
d. The Consortium will share the task fostering project results in regional contexts, supporting the exploitation approach with pilot cases and replication potentials in other industrial setting, pursuing the early involvement of stakeholders from other sectors of the process industry. To support this communication and dissemination activity, the Consortium will use supporting materials developed under the WP5 (5.2), such as project leaflets, brochure, poster, roll-up, infographic and video contents.

2. Intra-Industry Oriented dissemination and communication:

a. Expanding the number of industrial stakeholders involved in the process and validating the AI-CUBE business model for AI and BD implementation in process industry and required research needs on a larger scale. This phase will involve (approximately 30-50) additional stakeholders from European process industries.
b. In this phase, dissemination activities will focus on fine-tuning the knowledge base and research needs on the results of the project, making them available in a user-friendly mode, to a wider audience across different process industries.

3. Externally oriented dissemination and communication:

a. Expanding the audience of the AI-CUBE knowledge base and AI&BD technology implementation practices to a much wider audience across the process industries (50-100) and related stakeholders (500-3,000) throughout Europe. The AI-CUBE project will foster the take-up of its results, related pilot cases and roadmap for AI&BD technology implementation, targeting a vibrant interconnected European network of cross-sectoral process industries.
b. Dissemination activities will focus on further assessing the knowledge acquired in the AI-CUBE Project on the AI and BD technology level of maturity and implementation, and guaranteeing a wider audience of relevant stakeholders with the appropriate knowledge about the AI-CUBE Project and its results, in order to stimulate their take-up and acceleration of AI and BD deployment across different process industries in Europe.
c. Exploitation activities will be targeted to the development of a roadmap to secure a “sustainability strategy” for the AI-CUBE results on the implementation and penetration of AI and BD technologies across different process industries, leveraging potential cross-sectoral synergies. The project will investigate how the network of stakeholders active in AI and BD technology implementation can become a boosting element of AI and BD cross-industry diffusion beyond the H2020 project.

In order to properly execute and monitor the dissemination and exploitation activities, a detailed plan will be designed at the beginning of the AI-CUBE Project and updated regularly. This plan will include the more detailed definition of the deliverables and any additional activity required and will refine the approach to delivery outlined above. The following subsections present the preliminary dissemination, communication and exploitation activities for the AI-CUBE Project.

b) Communication activities

Communication related to promoting the AI-CUBE project and its results will be implemented through strategic and targeted measures for communicating and possibly engaging in a two-way exchange with the targeted groups. The AI-CUBE consortium will closely collaborate with SPIRE in its communication activities and with process industry experts identified per each sector. The target groups (i.e. 8 SPIRE process industry sectors, Value Chain partners of the European process industry, AI solution providers, and Policy Makers) will receive easily understandable information on technical project results, through user-friendly infographic and video contents. The communication plan will apply diverse communication regimes to different stakeholders, based on the importance they have for the project and their interest in the outcomes of AI-CUBE. Targeted group would be identified starting from industry sections that would be first AI-users:

1. Research
2. Management (C-suite, HR)
3. Operation
4. Logistics and VC managers

The initial phases of AI-CUBE project will only foresee generic news dissemination and communication, focusing on generating stakeholder awareness of the AI-CUBE Project and its potential role in assessing the process industry's needs for business model favouring the application and penetration of AI and BD technologies. In this phase, AI-CUBE will create a coherent identity for the project, supported with the design of a logo, leaflets, brochures, banners, posters, roll-up, infographic and video contents etc., and will deploy generic communication measures, such as the project website and periodic newsletters with generic information about the overall projects and related news on the EU process industry community. The AI-CUBE website will also offer general information, news & events, download and interactive sections, providing the opportunity to engage the audience over the duration of the project. The website will also be one of the means through which target audience will be invited to play an active role in the process of collecting, reporting, analysing, and disseminating content, e.g. blog/video-blogs, wikis, RSS, tagging and social bookmarking, music/photo/video sharing, podcasts. Such e-services will contribute to the involvement of end-users as active participants.

As the AI-CUBE Project progresses, the results will become available on the websites and more insights will be provided to the stakeholders that would be more attracted to the project activities. The information becoming available will provide the opportunity of involving additional external stakeholders into the project network by addressing their perceived interest in and attitude towards the project with targeted information. Strong media coverage will be another major target: media campaigns will be organized in relevant social networks to disseminate the AI-CUBE Project and raise awareness of the general public towards its key messages. In the end, scientific dissemination of project results will be pursued participating in international conferences, workshops, and seminars, and aiming for the publication of the project results and insights in academic journals, whose peer-review publication processes will further validate the scientific validity of AI-CUBE Project results. All publication will preferably have an open access status. For a more effective dissemination of the results, the AI-CUBE consortium will generate user-friendly infographic and video contents aiming at providing simplified visual graphics for the different process industries. These contents will include the CUBEs, visualising the current maturity and penetration levels of AI and BD technologies in different process industries and core processes; the AI-CUBE AI and BD Business Model Game, as a tool supporting industries in designing future business models; business cases that can potentially inspire AI and BD diffusion in process industry; and the roadmap providing detailed descriptions on how AI and BD should be better developed and can be taken-up.

c) Data Management and Management of IPR

Data and information collection, analysis, and dissemination will be crucial for the activities of the AI-CUBE Project. Therefore, the Consortium will adopt the following specific rules for data management throughout the process:

- Data generated during the projects and shared with the community will be validated as public information. If the results cannot be shared due to confidentiality issues, they will be presented in an adapted form for all targeted groups.
- Before any publication and/or wider use of the generated information, the content of the documents will be validated by the core group. This will contribute to ensuring the quality of the contents and the management of data confidentiality.
- Each strategy document and deliverable will be validated by the coordinator of the AI-CUBE Consortium, PNO, before the submission.
- AI-CUBE Project will manage a dedicated contact database and ensure that personal contact details will not be used elsewhere without specific agreement, thereby complying with the General Data Protection Regulation (GDPR). Contacts will be inserted in the communication database only after explicit agreement.
- Most of the data collected in the frame of the AI-CUBE project will be managed through its website, which will include a private and secured interface dedicated to the communication within the consortium partners.

Further details about the AI-CUBE data management, data protection, and confidentiality policies are available in Section 5 “Ethics and Security”.

3. Implementation

3.1 Work plan

The AI-CUBE project will be implemented through 3 key project phases:

1. **Phase 1: Set-Up** - Stakeholder engagement strategy, MLA methodology and initial technology assessment
2. **Phase 2: Gap analysis** - AI-CUBE business cases and gap analysis vs. current STATUS
3. **Phase 3: Roadmapping** - translate gap analysis into skills, data and RD&I requirements per SPIRE process industry sector
4. **Phase 4**: **Take-up** - Disseminate and promote the AI-CUBE roadmaps towards the wider process industries stakeholder communities

The activities of the project will be structured in 6 work packages (WPs) as shown in fig. 16.

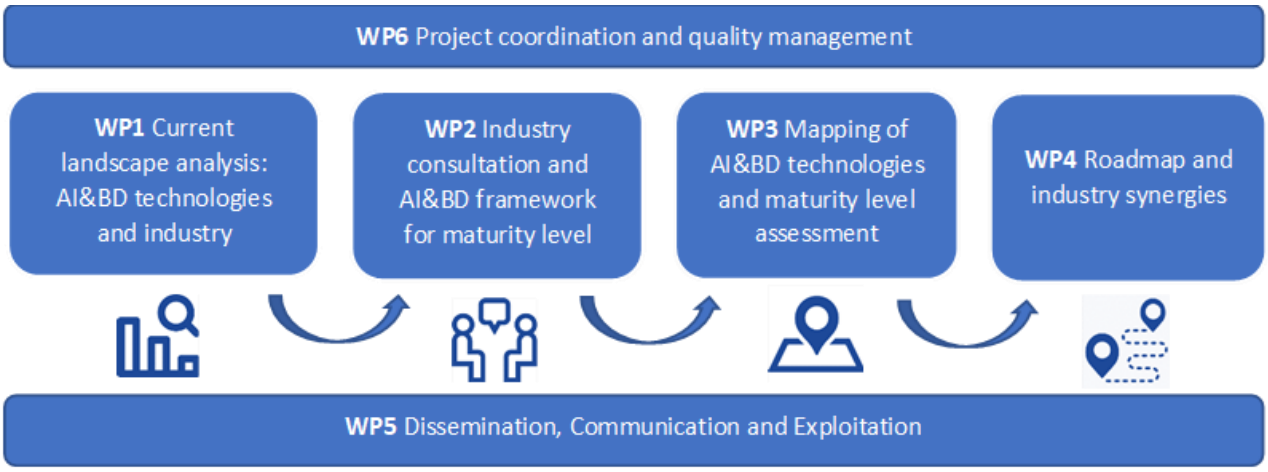


Figure 16. PERT Chart

In the **set-up phase 1** of the project, WP1 will start with an extensive literature and stakeholder analysis, to establish the current technological and industrial landscape regarding AI and BD technologies and process industry sectors in Europe, setting the basis of the following WPs work in defining mapping tools and the roadmap for AI and BD. In parallel, in WP2 a stakeholder engagement strategy will be set-up based on the MAMCA methodology, to work in a continuous way with the stakeholders and to design a framework to assess and categorize current Maturity Levels of AI and BD technologies' applications and deployment. **Phase 2** of the project will be targeted at carrying out a **gap-analysis**. First, WP3 will assess the maturity level of the previously identified technologies, while WP4 will define AI and BD driven Business Cases and Business Models for the process industries of the future. In WP4 a mapping of such visionary Business Cases and Models against the current STATUS (WP3) will be carried out to establish gaps and development needs per SPIRE process industry sector and per macro-application process areas (as defined in the CUBE). In **the roadmapping phase 3**, results from the previous phases will be translated into development needs and recommendations in terms of skills, data and RD&I requirements. As such WP4 will define a roadmap on AI and BD application in process industries for all the SPIRE industrial sectors. WP5 and WP6 will be horizontal to the whole project, focusing on dissemination and project coordination respectively, and culminating in the last **Take-up phase 4**, where the results of the project (the CUBEs, MLA framework, learnings and roadmaps) will be promoted towards the wider process industries communities (starting with > 100 industrial parks already identified in the previous INSPIRE project) and working in close collaboration with A.SPIRE.

Based on previous experience with SPIRE and FoF CSA's, the AI-CUBE consortium, aims for an early involvement of A.SPIRE in the project, already discussed with the A.SPIRE office. This will be leveraged to steer the project towards optimal and usable results, that can be taken-up by the eight (plus possibly pulp and paper, glass, etc.) process industries, as they come available in the course of the project.

TASK/WP	TASK/WP TITLE	TASK LEADER	START MONTH	END MONTH	DURATION IN MONTHS
WP1	**Current landscape analysis: AI & BD technologies and industry applications**	**IRIS**	**M1**	**M7**	**7**
T1.1	Identification of AI & BD technologies	IRIS	M1	M4	4
T1.2	Industrial processes analysis per sector and company size	IRIS	M3	M6	4
T1.3	Macro applications areas plan review and update	IRIS	M5	M7	3
WP2	**Industry consultation and AI&BD framework for maturity level**	**CNR**	**M5**	**M12**	**8**
T2.1	Stakeholders interviews and workshops	CNR	M5	M9	5
T2.2	Framework for the AI&BD technologies maturity level assessment	CNR	M6	M9	4
T2.3	Assessment of impact of AI&BD	CNR	M7	M12	6
WP3	**Mapping of AI & BD technologies and maturity level assessment**	**IML**	**M8**	**M18**	**11**
T3.1	Compilation of AI & BD technology per application and sector	IML	M8	M12	5
T3.2	Mapping AI & BD technologies per sector and macro area	IML	M11	M18	8
T3.3	Methods for the exploitation of AI methods	IML	M15	M17	3
T3.4	Industry validation	IML	M16	M19	3
WP4	**Roadmap and industry synergies**	**ZLC**	**M12**	**M24**	**10**
T4.1	Gaps and opportunities analysis – recommendations	ZLC	M15	M18	4
T4.2	CASE STUDIES – Business models	ZLC	M12	M23	8
T4.3	Development of the roadmap per sector and area	ZLC	M17	M24	8
T4.4	New paths of collaboration for ELSI	ZLC	M20	M24	5
T4.5	Validation	ZLC	M20	M24	5
WP5	**Dissemination, Communication and Exploitation**	**PNO**	**M1**	**M24**	**24**
T5.1	Definition of the strategy for dissemination of project results	PNO	M1	M3	3
T5.2	Communication activities	PNO	M1	M24	24
T5.3	Exploitation and dissemination	PNO	M1	M24	24
WP6	**Project coordination and quality management**	**PNO**	**M1**	**M24**	**24**
T6.1	Consortium and project administrative management	PNO	M1	M24	24
T6.2	Monitor technical Developments and Work Plan	PNO	M1	M24	24
T6.3	Quality Management	PNO	M1	M24	24

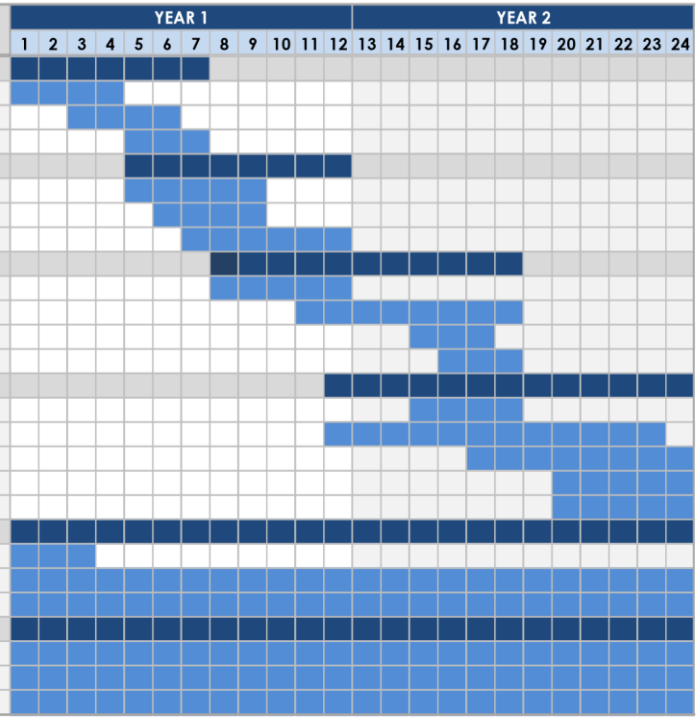


Figure 17. AI-CUBE Gantt chart

3.2 Management structure and procedures

The overall aim of the project management is to ensure a successful execution of the project to meet its objectives. To achieve this, the project manager and the partners will undertake all the necessary management and supporting actions to ensure high quality of its outcomes within the predefined time schedule and budget of the project.

3.2.1 Organisational Structure and Decision making

The organisational structure is characterised by the following constitutional elements (fig. 18):

The Coordinator
The Work Package and Task Leaders
The Contractual Partners forming the "Project Consortium and Decision Board"
The Experts forming the "AI-CUBE Expert Group"
The Stakeholders' Community forming the "AI-CUBE European Process Industry Network"

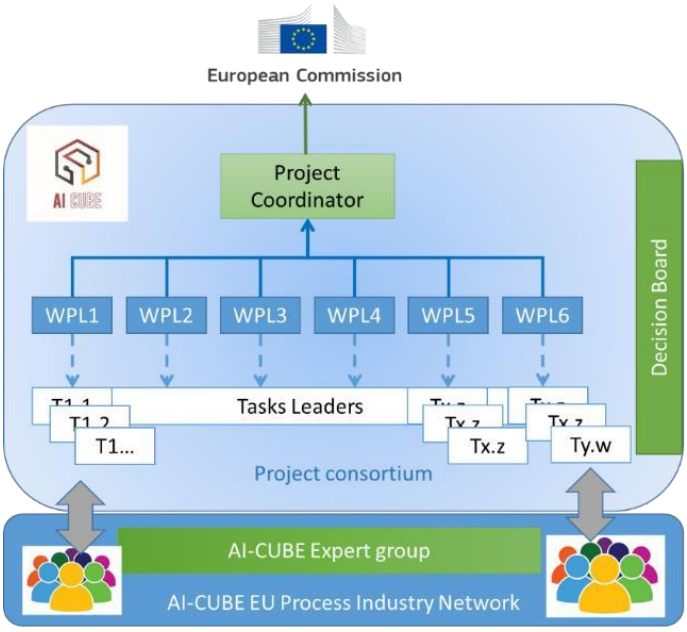


Figure 18. AI-CUBE Structural elements
Source: Own elaboration

The Coordinator

The coordinator will ensure a smooth and successful implementation of the project. The coordinator is responsible for the execution of the activities and the overall achievement of the project's objectives and thus for all contractual obligations towards the Commission. He may delegate specific subjects to the partners without however sharing partner's responsibility. The coordinating organisation provides the secretarial and organisational service for the project, including contractual and financial issues, reporting, the coordination of input to documents and their finalising, dissemination, or delivery, preparation of project meetings, the settingup and running of an infrastructure for internal and external communication, and running the every-day business of the project. The coordinator will report on regular basis to the EC (Project Officer, PO).

Project management and coordination will be carried out according to the highest quality, transparency and efficiency standards, and will be under the responsibility of PNO who will make use of its experience and expertise from the successful management of and participation in multiple EU funded projects, such as and the CSAs INSPIRE, NEXT-NET, New_Innonet and BioLinX, the RTD ROBOX, the RIA MEMERE, Indus3ES, in H2020, CSAs BIOTIC, R4R, CRM_INNONET, the IAs DESTINY and REE4EU and many others in 7FP. The coordinator will be Ronald Weerdmeester, European Innovation Manager, with more than 20 years' experience in managing EU funded projects and advisor to both CEFIC, Suschem and SPIRE. He will work in tandem with Chiara Eleonora De Marco, Innovation Consultant, and other sectorial experts within PNO.

The Work Package and Task Leaders

Work package and task leaders are contractual partners responsible for a work package or a task respectively. They have responsibility for the preparation and execution of the activities in their work packages or tasks according to the work plan. They may delegate specific sub-tasks to partners without however giving up the responsibility for the overall work package or task. Work package and task leaders are also responsible for

 providing general guidelines and interfacing with and for coordinating work with other work packages or tasks, in particular in meeting the deadlines for results delivery as these deliverables are needed as input for tasks to be timely performed in other work packages. They are also responsible for regular reporting on the work progress to the rest of the Project Consortium and Coordinator. In case of major developments that may impact the overall work plan, or problems in carrying out the tasks, they need to inform the Coordinator as soon as possible and discuss with him procedural and strategic issues in order to take common decisions to reach the project's objectives.

The Contractual Partners forming the "Project Consortium and Decision Board"

The contractual partners forming the project consortium encompass all staff from partner organisations involved in the project. They contribute to the achievement of the AI-CUBE CSA and accept that they are responsible for the tasks assigned to them according to the contract between the consortium and the European Commission. More specifically, they:

- participate in regular meetings,
- contribute to reports/deliverables,
- cooperate in elaborating procedures and the work to be carried out in the project,
- implement and carry out the activities as outlined in the work programme, mobilising the necessary resources and building up contacts with third parties as far as necessary,
- submit and review milestones and deliverables according to the responsibilities defined and agreed within the consortium and according to the quality assurance procedure,
- inform each other - and the coordinator, in particular - on all relevant events and activities related to the INSPIRE project,
- deliver cost statements and input to project activity and financial reports and provide information and data to the coordinator in order to ensure timely elaboration and submission of documents,
- collaborate and contribute to the elaboration of documents, tools and material, • carry out dissemination and promotion activities and represent the project exterally.

The consortium of AI-CUBE is small and involves only 5 partners. Together they will form the Project Decision Board, which is the actual decision making entity of the CSA; i.e. the place where procedural and strategic issues are discussed and decided in order to reach the project's objectives. Decisions may regard strategic, general and operational issues, e.g. the implementation of procedures and structures; the definition of activities to be carried out, the allocation and re-allocation of budget (in accordance with the EU contract); the management of conflicts and the monitoring of procedures and activities, and the inclusion of additional Associated Partners. One representative duly nominated by each partner organisation participates in the Project Decision Board. Should the nominated representative not be able to participate, he/she will nominate a deputy. Regular Project Decision Board meetings will be quarterly held physically or remotely - and according to necessity. Detailed decision-making processes within the consortium will be defined in the Consortium Agreement. Decisions will be taken by a simple majority vote. The voting will be open (non-secret), with one vote per partner. ***The Experts forming the "AI-CUBE Expert Group"***

This Expert Group will gather recognised experts in the various domains that are key to Europe's process industry and digital technology sector. They will participate in the AI-CUBE processes through consultation and validation processes and will be asked to review the related results and deliverables. They will also be given the opportunity to contribute to the results by delivering input to AI-CUBE activities of assessment and investigation of AI&BD technology deployment in European process industry. The experts will represent a broader community of specialists in Europe and beyond. International experts to be invited to the Expert Group and already identified within the individual networks of AI-CUBE partners are Mr. Jacques Komornicki, representing the European Technology Platform SusChem; and Prof. Da. Ir. Greet Van Eetvelde at the Ghent University, representing the SPIRE EPOS project (Enhanced energy and resource Efficiency and Performance in process industry Operations via onsite and cross-sectorial Symbiosis). Prof. Maria Chiara Carrozza, Member of the Task Force on Artificial Intelligence at the Agency for Digital Italy, Full Professor of Biomedical Engineering and Robotics at Scuola Superiore Sant'Anna, member of Member of the Steering Committee of the Quantum Tech FET Flagship, DG Communication Networks, Content and Technology. Emilia Garito, CEO at Quantum Leap IP, EU Innovation Expert for AI and Robotics. Dr. Francesca Lazzeri, Machine Learning Research and Applications at Microsoft (MA-USA). Robert Hoyle Brown, Vice President, Center for the Future of Work at Cognizant Technology Solutions (CA-USA). Prof. Roberto Zicari, ethical AI Goethe University Frankfurt.

Furthermore, as already anticipated to A.SPIRE representatives, in the desirable case AI-CUBE will result successful and receive EU funding, A.SPIRE will be invited to take part in the AI-CUBE Expert Group.

The Project Consortium will continuously work to expand this Expert Group with relevant representatives of the European process industries throughout the project. The Experts are involved in AI-CUBE activities but have no direct contractual obligations towards the EC. In return, they will not benefit from financial contributions except the reimbursement of travel expenses.

The Stakeholders' Community forming the "AI-CUBE European Process Industry Network"

This network extends the circle of contractual partners and experts and includes all relevant stakeholders. This includes all relevant actors from inside and outside the European process industries, starting from the members of the INSPIRE project list of 100 European industrial parks and related industrial stakeholders (process industries, users, and orchestrators applying AI&BD technologies to a minimum extent, and AI solution providers). The AI-CUBE stakeholder list will also tap into the European network created within the NEXT-NET project to reach a critical mass of relevant stakeholders from process industry, manufacturing, logistics and distribution. AI-CUBE European process Industry Network will iteratively test and expand AICUBE concepts eventually towards the wider European network of stakeholders, across the 8 SPIRE process industry sectors. The network will include international, European and national stakeholders representing the ICT community: users, technology suppliers, policy makers, SME associations, and research organisations. Moreover, from the Network will also contribute to the validation activities of the concepts and roadmap generated within the AI-CUBE project

Also, these stakeholders may build strong links to other relevant networks and stakeholder groups. As such they not only complement the expertise of the consortium, but also ensure the involvement of a wide stakeholder community, especially in the dissemination activities described in section 2.2. The Network will be engaged into the project activities by the participatory dissemination channels and through extensive networking activities, with the goal of planting the seeds for a network initiative that endures over the project duration and beyond.

3.2.2 Internal Quality Control

The detailed procedures of the quality management will be defined within the quality management plan (Task 6.3). In all elements of the work plan, at least 2 partners will contribute to the implementation of work packages and tasks, allowing an internal responsible/reviewer scheme. Each object (work package, task, deliverable, and milestone) is assigned to a responsible person in charge of the implementing the foreseen actions, producing the related deliverable and/or achieving the milestone. Reports will be reviewed by another project partner before evaluation by the coordinator. If the deliverable or milestone is rejected, the responsible partner will have to rework and resubmit the result. This workflow will be implemented through e-mail exchanges and videoconferences.

3.2.3 Operational planning

To ensure a smooth running of the activities to be carried out and ease the monitoring of the actions, the consortium will elaborate a detailed work plan based on the work plan submitted within this proposal, updating the tasks to be carried out in each work package, including the respective budgets and the division of labour among partners. The work plan will be supplemented by an updated time-schedule. This will be the essential planning and basic monitoring instrument of the project. The optimisation of resources to deliver the best value and larger impact will be targeted while driving the operational planning.

3.3 Consortium as a whole

The AI-CUBE consortium has 5 partners representing 3 EU countries (Italy, Spain, Germany). All partners in the consortium have well-defined tasks and responsibilities and their technical and nontechnical expertise corresponds well to the requirements of each of the WP tasks. The main advantage of the consortium is that it combines different expertise and past experiences that are fundamental to reach the goals and targets as well as established and expected impact.

The project will benefit from the work of each partner in related Innovation European and National Roadmaps (SPIRE, EFFRA, ALICE, etc) and this ensure the project will deliver a Roadmap which is compliant with the requirements derived also from the SPIRE Industry. Project partners have been directly involved in several research & development projects and among these the following ones related to AI and BD: SELIS (H2020MG-6.3-2015; N.690588), CONNECTED FACTORIES(H2020-FOF-11-2015, n. 723777), AGROinLOG

(H2020-RUR-08-2016; N.727961), FUTURING (H2020-NMBP-CSA-2016), DISRUPT (H2020-FOF-112016, n.723541), MOVAID (H2020-FOF10-2015, n.680754) and indirectly through network members in many others. Among these, project INSPIRE (SPIRE-06-2016; N.: 723748) have 3 partners of the consortium are involved and was committed to find new business models to strengthen the collaboration between process

industry and discrete manufacturing along supply chain collaboration. The link with INSPIRE is very strong and AI-CUBE can represent the follow-up of those activities.

On top of that, partners and project team have participated in coordination actions, CSA, such as: MOBILITY4EU, SENSE, SETRIS, WINN, SoCool@EU, RECREATE, CRM_INNONET, BIO-TIC, BioLinX, NEW_InnoNet, EXPLORE, focussed on identifying policies, establishing collaboration frameworks, identifying best practices and development of roadmaps. The team has a strong expertise in supply chain management (ZLC), production processes (CNR), industrial symbiosis (PNO), data driven technologies (IML and IRIS), and so on. This expertise will be complemented with the support of the Expert Group that will provide their expertise in other relevant projects.

The consortium has the ability to interface with the European Process Industry stakeholders. PNO is key partners to the European Technology Platform (ETP) SusChem and the European Council for the Chemical Industry (CEFIC). Both PNO and IRIS have strong ties with A.SPIRE, the SPIRE association. CNR-STIIMA is a major partners of the European Technological Platform on Future Manufacturing Technologies (MANUFUTURE) initiative and founding members of EFFRA. And ZLC promoted the European Technology Platform in Logistics, (ALICE) through the coordination of the EU-funded project WINN. IML are members of the Big Data Value Association (BDVA).

The Consortium can directly exploit the national networks of each partner as a way to get directly in contact with the national research and innovation activities already running in the thematic areas of the project, as well as to significantly increase results diffusion.

3.4 Resources to be committed

The duration of the AI-CUBE project will be 24 **months**. The total budget is **597.806,88 €** of which the Commission's contribution will be **597.806,88 €.** This consists in 444.745,50 € for personnel costs, 33.500 € for other direct costs, and 119.561,38 € to cover the indirect costs. Furthermore, a strong network shall be built that will be enabled to sustain over the project duration and beyond. Dedicated dissemination actions toward the European process industries and ICT community shall engage the all stakeholders and raise awareness.

All functions assigned in the project have gone through a thorough analysis of the prior experience and knowledge of each partner.

The budgeted costs are predominantly personnel costs and minor quantities have been assigned to cover costs for travel and accommodation and costs in relation with dissemination, communication and exploitation activities. Additionally, budget to cover travel expenses for members of the Expert Group has been allocated to the Coordinator's budget. Within the Coordinator's quota has also been allocated the budget for the AICUBE project website (€ 10,000), the creation of at least one video and infographic materials (€ 3,000) for dissemination purposes and the budget for Business Model Game workshop (€ 10,000).